

MSci Thesis:

Rigorous treatment of the Coulomb Potential in Non-Sequential Double

Ionization using the Coulomb Quantum-orbit Strong Field

Approximation

Tobin Holtmann

16009864

zcaptho@ucl.ac.uk

Supervisor: Prof. Carla Figueira De Morisson Faria

21 April 2020

Abstract

In this project we improve the existing model for the Recollisional Excitation with Subsequent Ionization (RESI) mechanism in Non-Sequential Double Ionization (NSDI) which is based on the Strong Field Approximation (SFA). The current model has been successful in predicting experimental results for RESI, but suffers from the same fundamental flaw as other SFA based models: the lack of the Coulomb potential in the continuum, which leads to a loss in dynamical behaviour. This model is improved by introducing the Coulomb Quantum-orbit Strong Field Approximation (CQSFA), a Coulomb corrected SFA model which has found success in High-Order Above Threshold Ionization (HATI), to certain elements of the mechanism. The result is a model that is part SFA, part CQSFA and temporarily named the 'Frankenstein' model. A momentum suppression along the axes of the Partial Momentum Distributions is identified as a new feature. An in-depth analysis is performed to explain this new feature.

Names of people should be written in capital letters.

Declaration of Authorship

I, Tobin Holtmann, confirm that the work presented in this thesis is my own work or the work of my project partner Gergely Eöry who collaborated on the same project. Any individual contributions are clearly indicated throughout this document. Information derived from other sources has clearly been labelled.

Contents

1	Introduction	4
1.1	Strong Field Physics	4
1.2	Non-Sequential Double Ionization	4
1.3	High Harmonic Generation and Above Threshold Ionization	5
1.4	Experiments, Coordinates and theoretical frameworks	6
1.5	Atomic Units and Useful Quantities	7
2	Theoretical Models	8
2.1	Numerical Quantum Ab-Initio Methods	8
2.2	Classical Models	9
2.3	Strong Field Approximation	9
3	SFA Theory	10
3.1	The Hamiltonian and Gauges	10
3.2	Volkov States	11
3.3	Dyson Integral Equation	11
3.4	The KFR Amplitude	12
3.5	Saddle Point Approximations	15
3.6	Quantum Trajectories and Orbits	17
3.7	Simple Man's Model	18
3.8	Complex Ionization time	19
3.9	Issues with the SFA	20
4	The Coulomb Quantum-Orbit Strong Field Approximation	21
4.1	Forward vs Backward Methods	21
4.2	CQSFA Derivation	22
4.3	The Tunnel-Exit Approximation	24
4.4	CQSFA Quantum Orbits	26
4.5	CQSFA Orbit Solver	28
5	The current NSDI SFA model	28
5.1	Subspace Projectors and Orthogonality	30
5.2	Solving NSDI orbits	32

5.3	The Uniform SPA	33
5.4	Momentum Constraints and Interference	34
6	The New 'Frankenstein' Model	37
6.1	The need for a new model	37
6.2	Aims and Objectives	38
6.3	Theory	38
6.4	Interpolation and Coding	41
7	Results, Analysis and Discussion	45
7.1	Partial Momentum Distributions	45
7.2	Initial Probability Distributions	46
7.3	Momentum-Mapping	49
8	Conclusion and Improvement	53
	Acknowledgement	54
	References	55

1 Introduction

1.1 Strong Field Physics

Non-Sequential Double Ionization (NSDI) is a correlated two electron process that forms part of Strong Field Physics and Attoscience. Strong Field Physics (SFP) is a research area of Atomic and Molecular Physics, in which individual atoms or molecules are trapped and bombarded with short bursts of strong lasers with intensities ranging from 10^{12} to 10^{16} Wcm^{-2} [1, 2] and very short wavelengths. In this SFP intensity regime, the strength of the laser's electric field and the atom or ion's binding potential on individual electrons can become comparable. The nuanced interplay between those forces leads to a whole variety of interesting processes like stimulated tunneling and multiphoton interactions that can happen within Attoseconds (one billionth of a billionth of a second) and manifest themselves in phenomena like Above Threshold Ionization (ATI) and High Harmonic Generation (HHG). The goal of SFP research is to fully understand and build models for these processes. This is not only interesting from a fundamental research perspective, as these processes play on the boundary of classical and quantum mechanical world, but also from a more practical one, since an in-depth understanding would allow us to harness these processes and potentially directly control chemical reactions. An example of this is Microsoft's Project Silica, which uses a femto-second lasers to etch data into a crystal at unprecedented storage densities [3]. Examples such as these already demonstrate the utility of accurately understanding such light and matter interactions.

One can go higher but then QED sets in.

not much restriction there anymore. They can be very or really short (see XFEL)

This is nice. Thanks for citing it!

1.2 Non-Sequential Double Ionization

Non-Sequential Double Ionization refers to the two electron process, in which the photoionization of one electron causes a second electron to also be removed from its parent ion or molecule, and can be found in several molecules and all noble gases [4–6]. It was first discovered in 1983, when the ionization rate in Xeon diverged significantly from the Sequential Ionization model (in which electrons are removed individually one by one) [1], giving rise to the now famous knee structure (fig. 1). The modern day explanation for this divergence is that one electron first enters the continuum¹ via tunnel ionization and is then driven back into its parent ion by strong laser field resulting in a collision that causes a second electron to also undergo ionization. This mechanism is known as the "three step model", which can also explain other SFP phenomena like HHG and Higher-order ATI (HATI) which are explained in the next subsection. In the three-step model:

1. An Electron enters the ~~into~~ the continuum through tunnel ionization.
2. The laser field accelerates the electron, which may change its direction back to the parent ion.
3. The electron either recombines and releases a photon (HHG), scatters elastically (high order ATI) or inelastically which ionizes a second electron (NSDI).

¹Formally the continuum refers the set of (finite) energy levels an electron can occupy in atom or ion, whose gaps in energy are negligible small that they effective become continuous. In this context it means the states in which an electron is no longer trapped inside an atom's or ions binding / coulomb potential

Nice to set that up. This definition is useful once you are dealing with⁴ structured continua, e.g.,

band gaps.

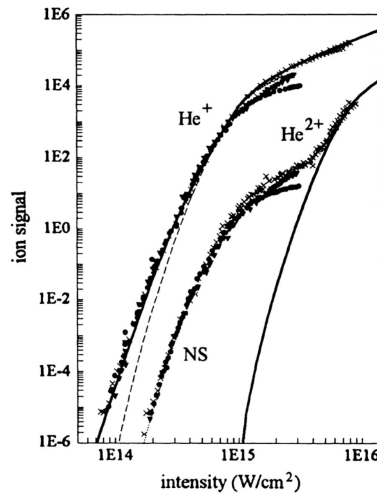


Figure 1: Early experimental measurements of Helium conducted by Walker et al. [2] which helped proved the existence of NSDI. The continuous line is the Sequential Ionization model that dominated at the time, and which diverges from experimental measurements below a threshold of $1.5 \times 10^{15} \text{ Wcm}^{-2}$. This shape is now widely known as the "Knee".

For NSDI, the three-step model can be subdivided into two further mechanisms known as Electron Impact (EI) ionization and Recollisional Excitation with Subsequent Ionization (RESI). In the EI the first electron has sufficient Kinetic energy to directly ionize electron 2 upon impact, whereas in RESI electron 1 only has enough energy to bring electron 2 into an excited state, which then after some time will tunnel ionize into the continuum. Originally EI was considered as the dominant mechanism, as it was simple to model classically and reliably predicted early experimental results [7–9] such as positive electron correlation (see fig. 2b for explanation) such as the so-called "V-shape" structure (fig. 2a). It was also believed that any quantum interference in RESI would be 'washed out' in experiments and would also exclusively lead to electron anti-correlation (see fig. 2b). Later experiments [10, 11] and theoretical models including the SFA [12, 13] showed that RESI leads to a cross-shape pattern which spreads out over the entire partial electron momentum distribution (see fig. 3).

It can still be, depending on the parameters range.

Excellent intro - to the point and very polished.

1.3 High Harmonic Generation and Above Threshold Ionization

Above Threshold Ionization is a multiphoton-process in which electrons are ionized with more photons than energetically required [14]. It was one of the earliest SFP processes discovered and has motivated many of the theoretical frameworks like the Strong Field Approximation (SFA) upon which also the current NSDI model discussed in this thesis is built. In direct ATI, the electron directly enters a detector upon ionization and is in contrast to the later discovered High Order ATI (HATI) in which an electron will reengage with the parent again, as described above. This results in new features such as holographic sidelobes and the 'spider' or fan-structure [15]. Although (H)ATI does not directly form part of this thesis' research, understanding it is important because the Coulomb Quantum-orbit SFA (CQSFA), the model with which we aim to improve the current NSDI SFA model, was developed for and extensively applied to (H)ATI. Another related process is High Harmonic Generation in which an illuminated ion source reemits light of energies of frequencies in integer multiples of the original light's frequency [16].

Totally agree and you make a great point

by explaining why this happens.

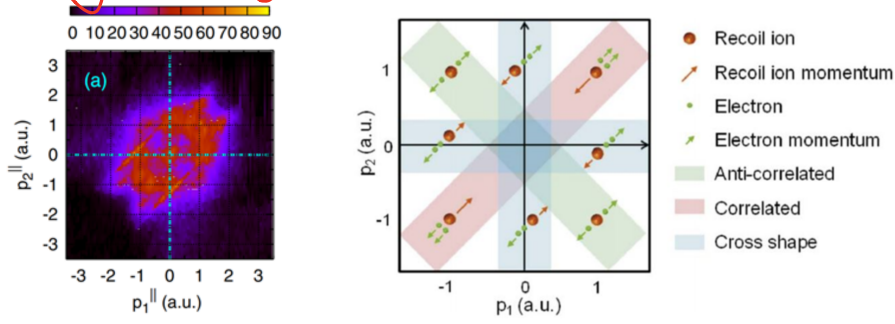


Figure 2: Panel (a) shows the PMD of the so-called 'V-shape' structure, which is a fingerprint of the electron-electron interaction [20]. This shape generally corresponds to positive correlation, which is predicted by the EI mechanism and classical models. Panel (b) explains signals in each quadrants of the PMDs are related to the different types of electron correlations. In (positive) correlation the two electron's momentum are aligned in the same direction which is generally predicted by EI. In anti-correlation the electron's momentum are aligned opposite to each other. It was originally thought that RESI leads exclusively electron anti-correlation, though this was later disputed. All values are in atomic units (a.u.) Sources for (a) and (b) are [21] and [22] respectively.

1.4 Experiments, Coordinates and theoretical frameworks of CPA.

NSDI like other SFP phenomena require strong lasers of intensities of up to 10^{12} Wcm^{-2} to 10^{16} Wcm^{-2} and ultra short periods (down to the femtosecond regime) [17]. Such requirements can only be achieved through special techniques such as Q-switching or Mode-locking [18], which pump out energy in very short bursts. Techniques like Cold-Target Recoil Ion Momentum Spectroscopy (COLTRIMS) [19] allow the photo-electron yield to be measured which is the intensity of (or the 'amount of') electrons entering the detector with a specific momentum, usually measured in components parallel and perpendicular to the laser field polarization. These photo-electron momentum yield graphs usually exhibit specific features like quantum interference, and are used as the main tool characterise SFP phenomena.

One general goal of SFP research is to develop physical models and deep understanding of the processes that cause these photo-electron distributions. Unfortunately many of these interactions happen on the timescale of attoseconds making physical measurements of 'what happens in between' difficult. This makes a good case for the need for the need of analytic theoretical models as these are the only tools through which we can develop a physical interpretation of these mechanisms that is intuitive to humans. To make theoretical models, we need to gauge the relevant quantities and make a few assumptions of the setup. For instance the photo-electron momentum spectrum is gauged through transition amplitude

$$M_{\mathbf{p}} = \lim_{t \rightarrow \infty, t_0 \rightarrow -\infty} \langle \psi_{\mathbf{p}}(t) | U(t, t_0) | \psi_g(t_0) \rangle, \quad (1)$$

whose absolute value squared $|M_{\mathbf{p}}|^2$ gives the probability of measuring an electron with momentum $\mathbf{p}$ at the detector. Here $U(t, t_0)$ is the time evolution operator of the system that links the systems initial ground state $|\psi_g\rangle$ at time t_0 to the final continuum state $|\psi_{\mathbf{p}}\rangle$ at the detector at time t .

We use a 3-D Cartesian coordinate system in which the origin of the system is located at the centre of the atom and assume that the laser's electromagnetic field is polarized along the z-axis, ie it is of the form $\mathbf{E}(\mathbf{r}, t) = \vec{e}_z E(\mathbf{r}, t)$. Since the detector is also located somewhere on along the positive z-axis,

This makes a good point, but you should also be aware of the parameter range.

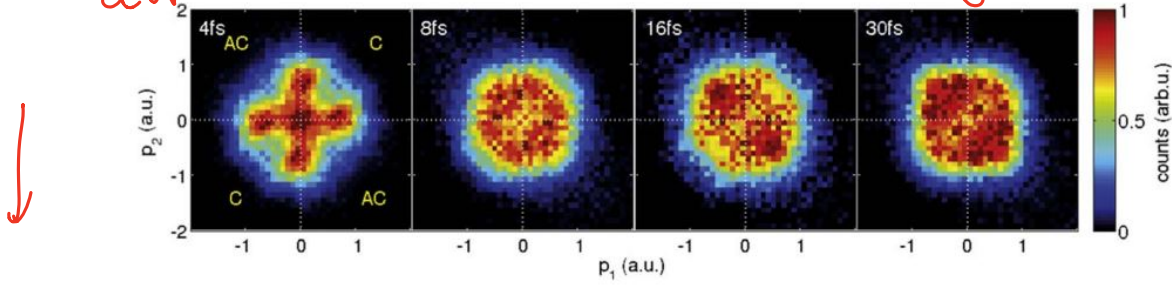


Figure 3: This diagram shows the PMD for the so-called 'cross shape' which over different laser cycles (one 2π phase of the laser's electromagnetic plane wave) spreads out over all quadrants of the PMD, with the final distribution having some preference for anti-correlation. This behaviour was also predicted by the RESI mechanism using methods such as the SFA. Since this signal covers all four quadrants, including the two quadrants associated with positive correlation, previous results attributed to EI could therefore now be explained by RESI, thus shifting the focus of research to RESI. Graph sourced from [22].

the positive z-axis is also referred to as the parallel direction $\vec{e}_z = \vec{e}_{\parallel}$. The system is also symmetric w.r.t rotations about this axis, which means that the exact orientation of the x and y-axis is arbitrary. Many quantities therefore only requires description with position or momentum in terms of parallel (e.g. $p_{\parallel}$) and perpendicular ($p_{\perp}$) components, effectively making this a 2-D Cartesian system. Formally $\mathbf{p}_{\perp} = p_{\perp} \vec{e}_{\perp} = p_x \vec{e}_x + p_y \vec{e}_y$ where $p_{\perp} = \sqrt{p_x^2 + p_y^2}$.

Another useful quantity is the electric field vector potential $\mathbf{A}(\mathbf{r}, t) = A(\mathbf{r}, t) \vec{e}_{\parallel} = - \int_{t_0}^t \mathbf{E}(\mathbf{r}, \tau) d\tau$, which can interpreted as the momentum transferred from the field to the electron per unit charge.

The explicit forms of $E(\mathbf{r}, t)$ for a realistic laser are typically complicated which makes theoretical predictions difficult. We therefore assume a simplified continuous, monochromatic and un-pulsed field in which the spatial components become negligent $E(\mathbf{r}, t) \approx E(t)$ as by the dipole approximation. The explicit form is $E(t) = E_0 \sin(\omega t)$, which gives a vector potential of $A(t) = \frac{E_0}{\omega} \cos(\omega t)$. For research of NSDI SFA with pulsed fields see MSci thesis [23].

Very critical for understanding for research in NSDI are the so-called Partial Momentum Distributions (PMDs), which are graphs displaying the expected photo-electron yield in terms of parallel momentum of electron 1 $p_{1\parallel}$ and electron 2 $p_{2\parallel}$ on the x and y axis respectively. These can be derived from the transition amplitude by removing out the parallel components through integration. The formal definition is

$$W(p_{1\parallel}, p_{2\parallel}) = \int |M(\mathbf{p}_1, \mathbf{p}_2)|^2 d^2 p_{1\perp} d^2 p_{2\perp} \quad (2)$$

1.5 Atomic Units and Useful Quantities

Unless explicitly stated, all equations and formulas throughout this document are in Hartree atomic units (abbr. as a.u.). This unit system sets the physical constants of the electron's mass m_e and charge e , the Reduced Planck Constant $\hbar$ and the inverse Coulomb Constant $4\pi\epsilon_0$ to unity. ie

$$m_e = e = \hbar = 4\pi\epsilon_0 = 1 \text{ a.u.} \quad (3)$$

As a result the speed of light in atomic units is given by $c = 1/\alpha \approx 1/137 \text{ a.u.}$, where α is the fine

structure constant. For more details and conversion table see [24]. This system of units is useful for Strong Field Physics (SFP) as the constants in many equations conveniently disappear. Furthermore Hartree units make numerical values more appropriate; for instance the period of single laser field cycle, and thus most interactions, frequently are on the order of femto seconds (10^{15} s), which in atomic units more appropriately is $\approx 41.1 \text{ a.u.}$ 10^{-15} s.

Another quantity particularly relevant to SFP is the ponderomotive energy, or cycle averaged quiver energy U_p . This is defined as the time averaged kinetic energy of a an electron in a (polarised) electric field. For a continuous uniform electric field of amplitude $E(t) = E_0 \sin(\omega t)$ and angular frequency ω , this becomes just the kinetic energy over one cycle:

$$U_p = \frac{1}{T} \int_0^T \frac{1}{2} A^2(t) dt = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} \frac{E_0^2}{2\omega^2} \sin^2(\omega t) dt = \frac{E_0^2}{4\omega^2} \quad (4)$$

as given here [25]. The ponderomotive energy becomes useful because it scales the amplitude and frequency of the field. Note that each 2π phase or each period of time $T = \frac{2\pi}{\omega}$ is known as a “cycle”. In most experiments involve exposure of only a few laser cycles.

2 Theoretical Models

There exist many different theoretical models or methods which can describe a variety of phenomena in SFP, and are broadly grouped into three categories: Ab-initio Quantum, Classical and Semi-classical analytical. This section briefly discusses their differences and strength and weaknesses.

Given that many processes like tunnel ionization are inherently non-classical it should be obvious that a Quantum Mechanical formulation is the most appropriate choice to model SFP phenomena. Therefore wave functions $|\psi(t)\rangle$ that fully solve the Time-Dependent Schrödinger Equation (TDSE) involving a Hamiltonian which fully takes the atomic potential V_c and the electron-field interaction V_{int} ,

$$H(\mathbf{r}, t) = \frac{\mathbf{p}^2}{2m} + V_c(\mathbf{r}) + V_{int}(t), \quad (5)$$

can accurately predict the vast majority of possible physics and dynamical behaviours involved. However finding fully analytical solutions to the TDSE without simplification or approximation is impossible in most cases. Hence there are three different methods of circumventing this issue.

For a comparison of these different methods see [26].

2.1 Numerical Quantum Ab-Initio Methods

Given that many processes like tunnel ionization are inherently non-classical it should be obvious that a Quantum Mechanical formulation is the the most accurate choice for the model SFP phenomena. Numerical ab-initio methods, such as the Qprop package [27], fully solve the TDSE using numerical methods from first principles. As this involves no simplification of the physics, they are able to model the electrons’ entire dynamical behaviour and as such have recreates many experimental findings, including the first evidences for RESI [28]. They however have two major downsides. For one, numerical calculations are very expensive requiring an extreme level of precision, as the strength of the interaction in SFP

phenomena can change by 100 eV within a fraction of a femtosecond [29]. However more problematic as a tool in research is the difficulty to extract a physical picture of the intermediate processes. For instance, in NSDI, distinguishing between RESI and EI is difficult since both processes are embedded in ab-initio calculations [30]. However, their ability to calculate predictions without loss of physics makes numerical TDSE a useful benchmark for analytical models. Another issue is that these methods require larger regions (or 'boxes') of numerical data-points for increasing wavelengths due to larger excursion amplitudes.

Here it is important to make the $1e^-$ - $2e^-$ distinction. Meanwhile it is relatively straightforward to solve the TDSE for $1e^-$ but it can get much harder for $2e^-$ s.

2.2 Classical Models

Models which significantly simplify the quantum mechanics to classical equations are usually referred to as "Classical Models". Any inherently non-classical effects are modelled through ad-hoc interventions. For instance in NSDI, the initial tunnel ionization step is modelled statistically as an ensemble of electrons where each one has a probability to be released into the continuum [31]. In the continuum; Newton's equation of motion (ordinary differential equations) govern the electron's dynamics, which are much easier to solve numerically than the partial differential equations of the TDSE [31]. However this great simplification comes with great disadvantages as classical theories have a hard time modelling more complex processes such as RESI where another ad-hoc procedure is required to the model tunnel ionization of electron 2. Another problem is that one also loses inherent quantum mechanical-features like the spreading of the wave packet and quantum interference [30]. The latter is especially problematic for RESI since the quantum interference between the different possible ionization channels (ie each possible excited atomic energy state of electron 2) largely determines the PMD pattern.

An important point here is that EI is much more

2.3 Strong Field Approximation

classical than RESI, if you wish.

The Strong Field Approximation (SFA) is a quantum-mechanical model that employs a series of intricately crafted approximations and simplifications to the transition amplitude (eqn. 1). It is a simple yet effective model and is widely employed throughout SFP research. Some even call it the "work-horse of SFP" [32]. Unlike classical theories, it is quantum mechanical in nature and excels at calculating quantum interference as well as the spreading of the wave-packet. Unlike numerical ab-initio models, it is mostly analytical, can model different processes independently which allowing intuitive physical interpretations of the intermediate processes to be extracted. The SFA developed gradually over time, and was influenced by many theories. Its main influence probably is Keldysh Ionization theory in the 1960s, which successfully predicted multiphoton processes in Hydrogen atoms [33].

It also follows a neat physical interpretation; it assumes that the electrons in their initial ground state are only affected by the binding potential as they are shielded away from the external field. After entering the continuum upon tunnel ionization, the electrons are assumed to only be affected by the laser's electric field neglecting the residual Coulomb binding potential. Although this originates mostly from the mathematical derivation (as detailed in section) this idea can also be justified from an intuitive perspective; in their ground state the electrons can be physically interpreted to be close atom centre, which means that the strength of the Coulomb binding potential V_c is strong and dominates over the

⊕ Difficult to say. I would definitely have

cited the other people.

external electric field. Since $V_c \propto -1/r$, it also means that Coulomb potential's strength rapidly depletes as the electron accelerates away from the core upon ionization. The ionization events also happen very quickly which gives the time-dependent external field little time to meaningfully change the electron's momentum before the majority of potential binding energy is transferred to the electron. After ionization in the continuum, the external field therefore dominates and forces the electron to oscillate as a plane-wave. See section 7.3 for further details of this intuitive interpretation.

Important: neglecting the Coulomb potential also simplifies things a lot, because the Volkov solution is analytically solvable.

3 SFA Theory

This section features a derivation of the theoretical framework of the Strong Field Approximation. Throughout this chapter, the Keldysh-Faisal-Reiss (KFR) amplitude of a direct process like ATI is used as an example to teach the main concepts and working principles of the SFA. The link between Feynman diagrams and transition amplitude expressions for higher order interactions is also explained. Furthermore the strengths and weaknesses as well as the need for improved models like the CQSFA should become apparent in this section. These concepts are all transferable and important to understanding the NSDI model.

3.1 The Hamiltonian and Gauges

Like with most quantum mechanical methods, a Hamiltonian that appropriately gauges the electron's dynamical behaviour is needed. The most general form employed is

$$H(t) = \frac{\mathbf{p}^2}{2} + V_c(\mathbf{r}) + V_{int}(t), \quad (6)$$

where V_{int} is the external field interaction operator, V_c the Coulomb / binding potential operator and $\mathbf{p}^2/2$ the kinetic energy operator. There is some freedom in the exact definition of these terms leading to two distinct gauge formulations.

In the **Length Gauge** the field interaction term is $V_{int}(\mathbf{r}, t) = \mathbf{r} \cdot \mathbf{E}(t)$ and the Coulomb potential is $V_c = -C/r$ where C is a strength constant (typically $0 \leq C \leq 1 \text{ a.u.}$) and r is magnitude of the electron's position vector $\mathbf{r}$. The Hamiltonian therefore becomes

$$H_L(t) = \frac{\mathbf{p}^2}{2} + \mathbf{r} \cdot \mathbf{E}(t) - \frac{C}{r}. \quad (7)$$

Applying the unitary Göppert-Mayer Gauge transformation $|\psi(t)\rangle \rightarrow e^{i\mathbf{r} \cdot \mathbf{A}(t)} |\psi(t)\rangle$ the TDSE ($i \frac{d}{dt} |\psi_L(t)\rangle = H_L |\psi_L(t)\rangle$) to the TDSE associated with this Hamiltonian ($i \frac{d}{dt} |\psi_L(t)\rangle = H_L |\psi_L(t)\rangle$) transform the Length Gauge into the **Velocity Gauge** [34]. Here the Hamiltonian has the form:

$$H_V(t) = \frac{1}{2} (\mathbf{p} + \mathbf{A}(t))^2 - \frac{C}{r}. \quad (8)$$

Note that the $\frac{1}{2} \mathbf{A}^2(t)$ term is sometimes ignored since it just represents a time-dependent phase factor in any observables [35]. In theory both gauges are equally valid and produce the same observables. However, following the approximation of the SFA, this is no longer the case. For reasons discussed in section 3.9, we will be using the length gauge from here on.

No citation to Faisal or Reiss in the previous section, so that it is difficult to understand this.

Don't forget to cite Volkov!

3.2 Volkov States

Vitally important to the SFA are the Volkov or Gordon-Volkov states, which physically represent the wave-function of a potential free electron oscillating as plane wave in an electric field. They were first derived in 1926 by W. Gordon [36]. Mathematically they are the solution to the TDSE of the field-only Hamiltonian

$$H_f(t) = H(t) - V_c(t) = \frac{1}{2}\mathbf{p}^2 + V_{int}(t) \quad (9)$$

without the presence of the Coulomb potential. By looking at the Velocity Gauge Hamiltonian (eqn. 8) one can almost guess why the Volkov states are given by

$$|\psi_{\mathbf{k}}^V(t)\rangle = |\mathbf{p} + \mathbf{A}(t)\rangle e^{-i \int_{t_0}^t [\mathbf{p} + \mathbf{A}(\tau)]^2 d\tau} = |\tilde{\mathbf{p}}(t)\rangle e^{-i \int_{t_0}^t \tilde{\mathbf{p}}^2(\tau) d\tau}. \quad (10)$$

Here $\mathbf{p}$ is the electron's drift momentum in the continuum (constant without the presence of Coulomb potential) and $\tilde{\mathbf{p}}(t) = \mathbf{p} + \mathbf{A}(t)$ is the classical canonical momentum which is sometimes also referred to as the field-dressed momentum. The $e^{-i \int_{t_0}^t \tilde{\mathbf{p}}^2(\tau) d\tau}$ term is typically called Volkov phase factor. For a full derivation of the Volkov states in a modern SFP context see section 4.2 of [35]. Associated with $|\psi_{\mathbf{k}}^V(t)\rangle$ is the Gordon-Volkov time evolution operator which maps the Volkov states from time $t' \rightarrow t$, which in the most general form is

$$U_V(t, t') = \int d^3k |\psi_{\mathbf{k}}^V(t)\rangle \langle \psi_{\mathbf{k}}^V(t')| \quad (11)$$

3.3 Dyson Integral Equation

Another vital background for the SFA derivation is the Dyson Integral Equation. This equation states that time evolution operator $U(t)$ for a wave function satisfying the TDSE with any Hamiltonian of the form $H(t) = H_0(t) + H_1(t)$ can be expressed as

$$U(t, t_0) = U_0(t, t_0) - i \int_{t_0}^t U(t, \tau) H_1(\tau) U_0(\tau, t_0) d\tau, \quad (12)$$

where U_0 is the time evolution operator satisfying the TDSE with Hamiltonian $H_0(t)$. The Dyson Equation is exact and is valid for any choice of H_0 and H_1 . It should be also noted that equation is also valid after swapping U and U_0 in the second term [37]. It can formally be derived by transforming the TDSE from the Schrödinger to the Interaction picture w.r.t. H_0 , re-expressing it as an integral and then transforming back into the Schrödinger picture. Using this and writing the Hamiltonian as $H(t) = H_f(t) + V_c$, the time evolution operator for the above system can be written as

$$U(t, t_0) = U_V(t, t_0) - i \int_{t_0}^t U(t, \tau) V_c(\tau) U_V(\tau, t_0) d\tau, \quad (13)$$

Alternatively we can also express the Hamiltonian as $H(t) = H_a(t) + V_c$ where $H_a(t) = \frac{1}{2}\mathbf{p}^2 + V_c(t)$ is the field-free atomic Hamiltonian which leads to

$$U(t, t_0) = U_a(t, t_0) - i \int_{t_0}^t U(t, \tau) V_{int}(\tau) U_a(\tau, t_0) d\tau. \quad (14)$$

Both of these expressions are equivalent and exact.

state and the continuum. (this is an important piece of information)

3.4 The KFR Amplitude

By substituting expression (14) for $U(t, t_0)$ in the general transition amplitude (eqn. 1) yields

$$\begin{aligned} M_{\mathbf{p}} &= \lim_{t \rightarrow \infty, t_0 \rightarrow -\infty} \langle \psi_{\mathbf{p}}(t) | U_a(t, t_0) | \psi_g(t_0) \rangle - i \int_{t_0}^t d\tau \langle \psi_{\mathbf{p}}(t) | U(t, \tau) V_{int}(\tau) U_a(\tau, t_0) | \psi_g(t_0) \rangle \\ &= -i \lim_{t \rightarrow \infty} \int_{-\infty}^t d\tau \langle \psi_{\mathbf{p}}(t) | U(t, \tau) V_{int}(\tau) | \psi_g(\tau) \rangle. \end{aligned} \quad (15)$$

Here property $U_a(\tau, t_0) | \psi_g(t_0) \rangle = | \psi_g(\tau) \rangle$ was used and the first term vanished due to orthogonality of the initial ground. Note that this equation is still exact. Now we will apply the formal SFA approximation itself. The SFA assumes approximates the final continuum state as Volkov states (ie assuming the Coulomb potential in the continuum can be neglected) and as a consequence replaces the general time evolution operator with the Volkov time evolution operator in the above expression. Formally

$$\begin{aligned} \langle \psi_{\mathbf{p}}(t) | &\xrightarrow{SFA} \langle \psi_{\mathbf{p}}^V(t) | \\ U(t, \tau) &\xrightarrow{SFA} U_V(t, \tau) \end{aligned} \quad (16)$$

which leads to

$$\begin{aligned} M_{\mathbf{p}} &\xrightarrow{SFA} -i \lim_{t \rightarrow \infty} \int_{-\infty}^t \langle \psi_{\mathbf{p}}^V(t) | U_V(t, \tau) V_{int}(\tau) | \psi_g(\tau) \rangle d\tau \\ &= -i \lim_{t \rightarrow \infty} \int_{-\infty}^t \langle \psi_{\mathbf{p}}^V(\tau) | V_{int}(\tau) | \psi_g(\tau) \rangle d\tau. \end{aligned} \quad (17)$$

Note that this ~~is~~ equation remains valid by replacing V_{int} with V_c which is described in chapter 3 of [30] and also in [37], which gives

$$M_{\mathbf{p}}^{KFR} = -i \lim_{t \rightarrow \infty} \int_{-\infty}^t \langle \psi_{\mathbf{p}}^V(\tau) | V_c(\tau) | \psi_g(\tau) \rangle d\tau. \quad (18)$$

Equations (18) and (17) are known as the KFR amplitudes and encapsulate the physical interpretation of the SFA; the electron remains in its ground state until a time τ , when ionization happens effectively instantly and afterwards the electron oscillates as plane wave in the electromagnetic field.

The original non-approximated transition amplitude (eqn. 1) technically describes any and all valid interactions where the electron starts out in the ground state and ends up in the detector momentum $M_{\mathbf{p}}$. This includes any higher processes where the electron intermediately re-engages with the original parent ion. The KFR amplitudes only describes 1st order (direct) ionization process like ATI, where upon ionization the electron uneventfully travels towards the detector. This is because the true continuum states are replaced with Volkov states in which the drift momentum is conserved (the canonical momentum is $\tilde{\mathbf{p}}(t) = \mathbf{p} + \mathbf{A}(t)$). This is obviously a reduction in dynamical behaviour but also means that further work is needed to model re-collision and scattering which are essential to higher order processes like NSDI, HHG and HATI. To model re-collisions within the SFA, one must first substitute (14) for $U(t, \tau)$

in equation (13) and then re-apply the same procedure to replace $V_{int} \rightarrow V_c$, which yields

$$\begin{aligned}
 M_{\mathbf{p}}^{(1,2)} &= (-i)^1 \lim_{t, t_0 \rightarrow \pm\infty} \int_{t_0}^t dt_1 \langle \psi_{\mathbf{p}}^V(t) | U_V(t, t_1) V_c(t_1) U_V(t_1, t_0) | \psi_g(t_0) \rangle + \\
 &\quad (-i)^2 \lim_{t, t_0 \rightarrow \pm\infty} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \langle \psi_{\mathbf{p}}^V(t) | U_V(t, t_2) V_c(t_2) U_V(t_2, t_1) V_c(t_1) U_a(t_1, t_0) | \psi_g(t_0) \rangle \\
 &= (-i)^1 \lim_{t \rightarrow \infty} \int_{-\infty}^t dt_1 \langle \psi_{\mathbf{p}}^V(t_1) | V_c(t_1) | \psi_g(t_1) \rangle + \\
 &\quad (-i)^2 \lim_{t \rightarrow \infty} \int_{-\infty}^t dt_1 \int_{-\infty}^{t_1} dt_2 \langle \psi_{\mathbf{p}}^V(t_2) | V_c(t_2) U_V(t_2, t_1) V_c(t_1) | \psi_g(t_1) \rangle.
 \end{aligned} \tag{19}$$

One can see that the first term is the KFR amplitude (1s order direct) and the second term is the re-collisional term. The 1, 2 hyper-script indicates that this expression contains both first and second order terms. It should be easy to see that repeatedly following similar procedures allows one to generate a formula for processes up to n th order interactions, constituting to as field dressed Born series. The KFR amplitude therefore only represents a zeroth order Born series approximations [15].

Again it is possible to extract a neat physical intuition for physical interpretation from the recollision expression of the second term; the electron is initially in its ground state $|\psi_g(t_0)\rangle$ until it gets ionized at time t_1 , after which it will be in the continuum (approximated as plane wave Volkov states). At time t_2 it scatters with the atom (indicated through binding potential term $V_c(t_2)$) after which it stays continuum (again approximated as plane wave) until it reaches the detector with drift momentum $\mathbf{p}$ at time t .

It is also possible to draw Feynman diagrams to model these interactions. Feynman diagrams visualise an interaction in a capacity that is intuitive to humans but also have strict mathematical meaning as they are associated with specific transition amplitudes. This is useful because in order to the transition derive transition amplitude expressions for a processes, we can simply consider the Feynman diagram which is much more intuitive to humans. To do this, we model simply any ionization or atomic scattering event as vertex in the diagram with a V_c term in the transition amplitude expression. Any line connecting two interaction vertices in the diagram corresponds to the application of a appropriate time-evolution operator (or relevant approximation if necessary) in transition amplitude. Examples of this are given in fig. 4. In these diagrams $dM_{\mathbf{p}}^{(1)}(t_0, t_1, t)$ and $dM_{\mathbf{p}}^{(2)}(t_0, t_1, t_2, t)$ indicate first order (direct) and second order (re-scattering) processes which happen at the specific event times indicated the brackets. To get the full transition amplitudes $M_{\mathbf{p}}^{(1)}$ and $M_{\mathbf{p}}^{(2)}$ we must consider the interactions at every possible event time, which simply means integrating these terms over all possible interaction times. This yields the expressions 19 as required. The causality condition $\text{Re}(t_0) \leq \text{Re}(t_1) \leq \text{Re}(t_2) \leq \text{Re}(t)$ is required and prevents these interactions from diverging. Note that these interaction diagrams are not true Feynman diagrams and should be seen more as a general expression of the quantum mechanics. Whilst the horizontal axis represents time, the vertical axis does not always strictly represent position or momentum. Technically any higher order expression in the Born series can be considered, but these processes become increasingly less probable and thus can be neglected.

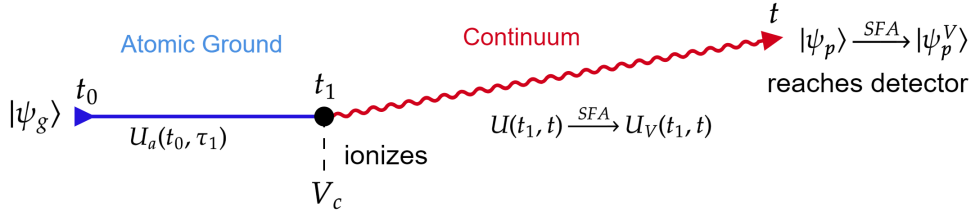
This demonstrates one of the biggest strengths of semi-analytical models like the SFA. Through picking the correct term in the Born series, we can study the likelihood and behaviour of any specific interaction and ignore others. For instance in the above example, we could just calculate and study the second order re-scattering term and ignore the first order (or vice versa) with the SFA. This is not possible, for

(X) I love this. Are you preparing for when the machines take over? sorry, I couldn't resist :)

Totally agree - it is really helpful when things are 'clear-cut'.

a) 1st Order (Direct)

$$dM_p^{(1)}(t_0, t_1, t) = \langle \psi_p^V(t) | U_V(t, t_1) V_c(t_1) U_a(t_1, t_0) | \psi_g(t_0) \rangle$$



b) 2nd Order (Rescattering)

$$dM_p^{(2)}(t_0, t_1, t_2, t) = \langle \psi_p^V(t) | U_V(t, t_2) V_c(t_2) U_V(t_2, t_1) V_c(t_1) U_a(t_1, t_0) | \psi_g(t_0) \rangle dt_1 dt_2$$

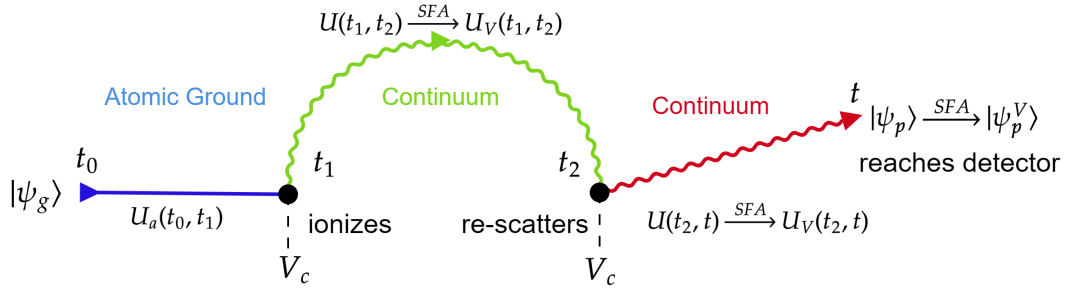


Figure 4: Feynman like interaction diagrams for a 1st and 2nd order ionization process. The purpose of this diagram is to visually explain the substitutions of wave states and time evolution operators in the transition amplitudes with the SFA, and also to highlight how the transition amplitude is related to Feynman diagram. The gray dashed lines indicate which operator in transition amplitude is considered to model each interaction. Any time evolution operator or bra-ket state shown is exact and accurate unless explicitly indicated to be approximates by the SFA through arrow $U_V(t_1, t_2)$. Here dM_p indicate the simplified integrands after the SFA is applied and which need to be integrated over their respective event times.

example, with ^aAb-initio numerical TDSE method as both first and second order processes are equally valid; both have same initial ground state and final momentum state at detector and thus are equally incorporated into the general TDSE solution. This makes it very difficult to distinguish between them. This makes the SFA a great tool in research to develop a human compatible understanding of these processes.

3.5 Saddle Point Approximations

The KFR amplitude (eqn. 18) among other SFA based transition amplitude expressions include integrals that are not easy to evaluate analytically. This may become especially apparent when considering forms of the electric field that are more complicated than $E(t) = E_0 \sin(\omega t)$. Therefore numerical evaluation or further analytic simplification is needed to progress further. A commonly used method is the so-called Saddle Point Approximation (SPA) also known as the ⁶Method of Steepest Descent". The SPA is applicable to multivariable integrals as well as path integrals in functional form. To best convey its working principle I will here derive it only for one variable and will use the KFR amplitude (18) as an example.

First step however is to bring the transition amplitude into a more explicit format that is applicable to the SPA. This is done by substituting the explicit form $|\psi_g(t_1)\rangle = |\psi_g\rangle e^{-iI_p t_1}$ of the ground state (where I_p is the ionization potential) and the explicit form of the Volkov phase (with the upper limit applied in the integral) into the KFR amplitude (17 which yields

$$M_{\mathbf{p}}^{KFR} = \int_{-\infty}^{\infty} dt_1 \langle \tilde{\mathbf{p}}(t_1) | V_c(t_1) | \psi_g \rangle e^{-i(I_p t_1 + \int^{t_1} \tilde{\mathbf{p}}^2(\tau) d\tau)} = \int_{-\infty}^{\infty} dt_1 V_{cg}(t_1) e^{-iS(t_1)}. \quad (20)$$

Here the term $V_{cg}(t_1) = \langle \mathbf{p} + \mathbf{A}(t_1) | V_c(t_1) | \psi_g \rangle$ is known as the continuum-ground prefactor. Note that the time t_0 in the ground state and in the lower limit of the Volkov phase is dropped but implied. The exact form of prefactor $V_{cg}(t_1)$ for now is ~~really~~ not really important; all that needs to be known is that it varies slowly w.r.t t_1 compared the exponential term $e^{-iS(t_1)}$.

The above expression (20) is an integral of the general form

$$I(\lambda) = \int_{-\infty}^{\infty} dx P(x) e^{\lambda S(x)} \quad (21)$$

which is applicable to the single-variable SPA. Here parameter λ should be sufficiently large such that $P(x)$ varies slowly w.r.t. x compared the exponential term $e^{\lambda S(x)}$. In our example above $\lambda = -i$ which is sufficient. This means that the overall shape of the integrand ^{is} is largely determined by the behaviour of $S(x)$. Due to the nature of exponentials, the $e^{\lambda S(x)}$ ^{turns} quickly becomes vertical unless $S(x)$ is horizontal, which will be the case if $S(x)$ has a saddle point, e.g. at $x = x_s$. The condition for a saddle point is

$$\left. \frac{dS(x)}{dx} \right|_{x=x_s} = 0. \quad (22)$$

Since a vertical line has little area under, this means that the majority of the integral will come from a saddle point as visually demonstrated in fig. 5. This can be exploited by approximating the integrand's shape using two appropriately shaped Gaussian bell curves that are overlayed at the saddle point as shown in the figure. We essentially have replaced a general function $e^S(x)$ of potentially non-existent

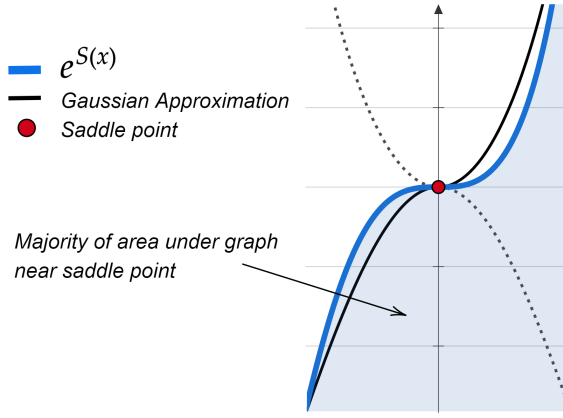


Figure 5: Visual demonstration of the SPA. The thick blue line shows $e^{S(x)}$, which becomes horizontal around the Saddle point of $S(x)$ (red dot), and quickly becomes vertical away from the saddle point. The black thin line shows the Gaussian approximation with which the general function is approximated. The grey dotted lines are the remaining parts of the two Gaussian bell curves.

derivative, with a function whose integral we definitively do know. The Gaussian shape can be motivated by considering a 2nd Order Taylor expansion $S(x) \approx S(x_s) + \frac{1}{2}S''(x_s)(x - x_s)^2$ where the 1st order term vanishes because of the saddle point condition 22. Substituting this into the the integral yields

$$\begin{aligned}
 I(\lambda) &\approx \int_{-\infty}^{+\infty} dx P(x_s) e^{S(x_s)} e^{\frac{\lambda}{2} S''(x_s)(x-x_s)^2} \\
 &= P(x_s) e^{S(x_s)} \int_{-\infty}^{+\infty} dx e^{\frac{\lambda}{2} S''(x_s)(x-x_s)^2} \\
 &= \sqrt{\frac{2\pi}{\lambda S''(x_s)}} P(x_s) e^{S(x_s)}.
 \end{aligned} \tag{23}$$

For this to appropriately work every saddle point $x = x_s$ must be considered. The final expression of the single-variable SPA therefore is

$$I(\lambda) \approx \sum_s \sqrt{\frac{2\pi}{\lambda S''(x_s)}} P(x_s) e^{\lambda S(x_s)}. \tag{24}$$

For the example of the KFR amplitude, the Saddle Point Equation is

$$I_p + [\mathbf{p} + \mathbf{A}(t_1)]^2 = 0, \tag{25}$$

which must be solved to find the saddle point times t_{1s} and finally simplifies the amplitude to

$$M_{\mathbf{p}}^{KFR} \approx -i \sum_s V_{cg}(t_{1s}) \sqrt{\frac{2\pi i}{S''(t_s)}} e^{i(I_p t_{1s} - \int^{t_{1s}} \tilde{\mathbf{p}}^2(\tau) d\tau)}. \tag{26}$$

With the SFA and SPA, we effectively have made evaluating transition amplitudes as easy as finding the saddle points of a single equation. However there is a caveat: the saddle points must be well separated, as otherwise the Gaussian bell curves start overlapping, meaning that the original function is not reasonably approximated anymore. However for this scenario there is the Uniform Saddle Point Approximation that takes closely located saddle points into account and which has to be used for the NSDI model. For an in-depth derivation of the SPA see [38].

By conceptually doing the same steps as above, the SPA can be extended to a integral of N variables of the form

$$I_N(\lambda) = \prod_{n=1}^N \left(\int_{-\infty}^{\infty} dx_n \right) P(x_1, \dots, x_N) e^{\lambda S(x_1, \dots, x_N)} \quad (27)$$

$$\approx \sum_s \left(\frac{2\pi}{\lambda} \right)^{N/2} \frac{P(x_{1s}, \dots, x_{Ns})}{\sqrt{\det(S''(x_{1s}, \dots, x_{Ns}))}} e^{\lambda S((x_{1s}, \dots, x_{Ns}))}.$$

Here $S''(x_1, \dots, x_N)$ is the Hessian matrix of 2nd derivatives of the action, which explicitly is:

$$S''(x_1, \dots, x_N) = \begin{pmatrix} \frac{\partial^2 S}{\partial x_1 \partial x_1} & \frac{\partial^2 S}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 S}{\partial x_1 \partial x_N} \\ \frac{\partial^2 S}{\partial x_2 \partial x_1} & \frac{\partial^2 S}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 S}{\partial x_2 \partial x_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 S}{\partial x_N \partial x_1} & \frac{\partial^2 S}{\partial x_N \partial x_2} & \cdots & \frac{\partial^2 S}{\partial x_N \partial x_N} \end{pmatrix} \quad (28)$$

The saddle point condition for each saddle point s is

$$\frac{dS}{dx_{1s}} = \frac{dS}{dx_{2s}} = \dots = \frac{dS}{dx_{Ns}} = 0. \quad (29)$$

This is used in the SFA NSDI model later and is derived in [18]. It is possible to even go much further than this and extend the SPA to path integrals and functionals of the form

$$I(\lambda) = \int \mathcal{D}x F[x] e^{\lambda S[x]} \quad (30)$$

$$\approx \sum_s \frac{F[x_s]}{\sqrt{\det S''[x_s]}} e^{\lambda S[x_s]}.$$

This theory is known the Van-Vleck ~~method~~ ^{Gutzwiller} method [39]. Here $S''[x_s]$ is the path integral functional form of the Hessian and unfortunately is quite difficult to evaluate because and may depend on the variables on which the functions x act. Given the right conditions this term can be simplified to differential operators. The saddle point equations can also be expressed in terms of path integrals. This is used for the CQSFA but path integrals and functionals are beyond the scope of this project and thesis. Any such expression shown in thesis is directly quoted from another source. The above equation was simply included to show that the working principle of the SPA can applied to a variety of different mathematical formalism.

we should have a look at them afterwards.

3.6 Quantum Trajectories and Orbits

We call the times solutions t_s to the saddle point equations like eqn. (25) the 'ionization times', ie the times at which an electron will enter into the continuum. Technically an electron can enter the continuum at any time it wants but there are different probabilities of doing so. By the logic of SPA, the times t_s that satisfy saddle point equations, are the times at which ionization is most likely, and hence we assume that ionization happens only at these 'ionization times'. Each of these ionization times yields a somewhat unique trajectory through phase space that the electron will take to the detector. These trajectories are collectively referred to as "Quantum Trajectories". For the SFA it is possible to categorize the trajectories into two different sets, each known as a "Quantum Orbit". This becomes apparent when looking back

the concept is more general and typically

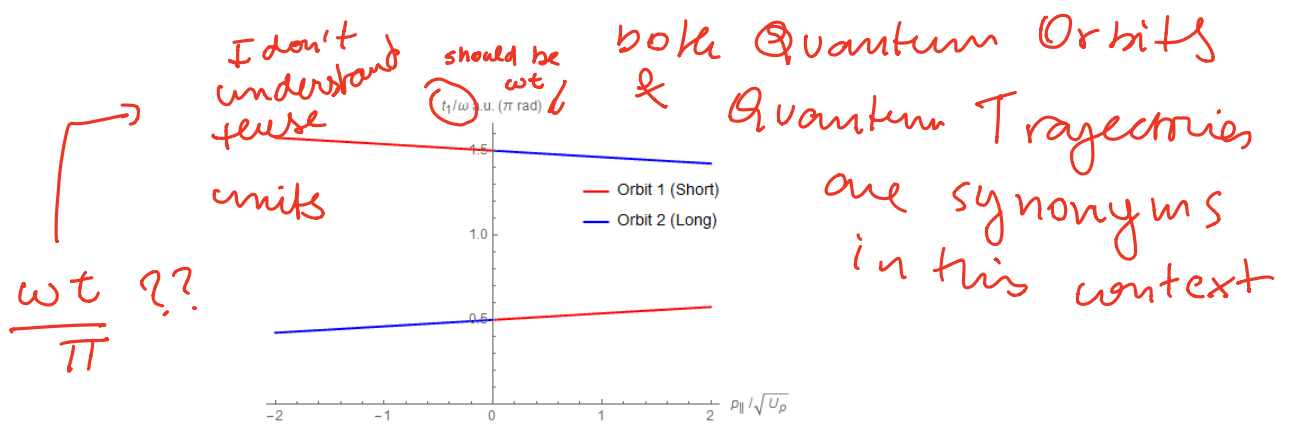


Figure 6: All possible solutions of saddle point equation 31 for the KFR amplitude as a function of parallel momentum $p_{\parallel}$ and how they are divided up into Orbit 1 / short (red) and Orbit 2 / long (blue). This is at zero perpendicular momentum $p_{\perp} = 0$, and wavelength $\omega = 800\text{nm}$ and intensity $I = 10^{13}\text{W/cm}$. This are the solution for one cycle only. Shifting this this solution by 2π gives the next solutions. Note how each orbit is discontinuous at $p_{\parallel} = 0$.

at

the KFR amplitude example. The full set of solutions to saddle point equation (25) is

$$t_{1s} = \frac{2\pi n}{\omega} \pm \frac{1}{\omega} \arccos \left(\frac{-p_{\parallel} \mp i\sqrt{2I_p + p_{\perp}^2}}{2\sqrt{U_p}} \right). \quad (31)$$

Here n is an integer and the term $\frac{2\pi n}{\omega}$ just shifts the solution by each laser cycle. The choice of $\pm$ therefore leaves behind two unique solutions within each cycle, which we can define the following:

Orbit 1 or “Short”:

$$t_{1s}^{\text{short}} = \frac{1}{\omega} \arccos \left(\frac{-p_{\parallel} - i\sqrt{2I_p + p_{\perp}^2}}{2\sqrt{U_p}} \right) \quad (32)$$

and Orbit 2 or “Long”:

$$t_{1s}^{\text{long}} = \frac{2\pi}{\omega} - \frac{1}{\omega} \arccos \left(\frac{-p_{\parallel} + i\sqrt{2I_p + p_{\perp}^2}}{2\sqrt{U_p}} \right). \quad (33)$$

Please note that these are the definitions for positive perpendicular momentum $p_{\perp} > 0$ only. For negative momentum $p_{\perp} < 0$ the definitions for orbit 1 and 2 switch in the above equations, meaning that each orbit is discontinuous at $p_{\parallel} = 0$ as shown in figure 6. Defining it this way may seem weird at first, however, but has physical significance. One can notice that within each π (half cycle) orbit 1’s ionization times are always larger than those of orbit 2, meaning that electrons with orbit 2 trajectories take spend a longer time in the continuum to reach the detector. The physical interpretation is that the electrons in orbit 1 (short) enter the continuum on the side of the atom that is closer to the detector whereas in orbit 2 (long) the electrons enter the continuum on the opposite side. With some Coulomb corrected SFA theories like the CQSFA, these two Orbit definitions are extended to two further orbit types, as shall be explained later.

3.7 Simple Man’s Model

Looking at figure 6, one can notice that Orbit 1 and 2 are centered around $\frac{\pi}{2}$ radians at zero parallel momentum. This is actually the case for any set of field parameters (ω and E_0) and I_p . This happens to be also the time when the field $E(t) = E_0 \sin(\omega t)$ is at its maximum. Therefore ionization happens at

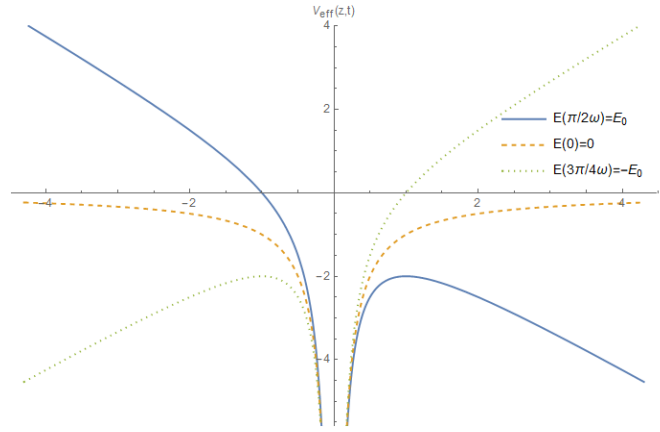


Figure 7: The effective potential formed by the Coulomb potential and the laser field, showing how the barrier is affected when the laser field $E(t)$ is maximum (blue), minimum (green dotted) and at a crossing (orange striped). At the crossing, the the effective potential is identical to the original atom's coulomb potential.

those times when the field has reached it's maximum or minimum. This is a general result known as the Simple Man's Model [40] which can be used as a general rule of thumb for many different processes. It also has a good physical intuition; ignoring the perpendicular components, an electron is classically only able to escape the potential well formed by the Coulomb potential $V_c(z) = -\frac{C}{z}$ if its kinetic energy is larger than the potential barrier. Quantum-mechanically an electron can still escape through the potential barrier via tunneling. However the probability generally diminishes the larger and wider the barrier gets. The addition of electromagnetic laser field $E(t) = E_0 \sin(\omega t)$ gives a dipole potential of $V_{field}(x, t) = xE_0 \sin(\omega t)$ that together with the Coulomb potential forms an effective time and space dependent potential of

$$V_{eff}(z, t) = V_c(z) + V_{dipole}(z, t) = -\frac{C}{z} + zE_0 \sin(\omega t). \quad (34)$$

When the field is maximum or minimum, the effective potential barrier well is lowered and and opens up on one of its sides therefore is 'the most classical'. Hence an electron should have the highest probability of escaping at this time. Stretching this intuition a bit further, one could argue that for 2nd order processes an electron should have highest probability to re-collide when the field as it crossing (ie equal to zero) as the potential barrier is equally closed on both sides. One can think of this 'its the easiest for the electron to bump into the barrier' and hence interact with the the atom, ie recollide.

Using the simple man's model's prediction of ionization at the maximum and re-collisions at crossing as a 'sledgehammer' for solving SFP problems is dangerous because it is only approximate and doesn't always tell the entire story. However it can be used for initial guesses when the saddle point ionization are more complicated and have to solved numerically using iterative solvers, as will be the case in the NSDI model.

3.8 Complex Ionization time

As one can see from eqn. (25), saddle point equations within the SFA model, typically allow for complex solutions for the ionization time. Imaginary time may appear non-physical at first, however ~~from a~~

SPM is something else - is when you consider classical trap & neglect the SFA & IP-10

This assertion is great and shows you really understand how to solve the problem.

quantum mechanical it does make sense within the SFA model perspective. This again is best explained through an example. Looking at eqn. (26), the SPA solution for the KFR amplitude, one can notice a $e^{iI_p t_{1s}}$ factor. For a complex ionization time $t_{1s} = \Re t_{1s} + i\Im t_{1s}$ this becomes $e^{I_p \Im t_{1s}} \times e^{iI_p \Re t_{1s}}$. The imaginary time acts as a scaling factor for the transition amplitude due to which larger imaginary times therefore constitute to a greater likely-hood for the tunneling process. Thus we can interpret the concept of non-classical imaginary time as an indicator for the non-classical concept of tunneling. One should note that the real parts still follow our classical understanding of time, which therefore demands the causality for interaction events and is commonly expressed as $\Re t_0 \leq \Re t_1 \leq \Re t$.

This is a really nice explanation. Very clear and helpful!

3.9 Issues with the SFA

Despite the SFA's many strengths, it does have a few flaws and weaknesses. Almost all of them can be attributed to SFA's main assumption: the lack of the Coulomb potential in the continuum.

One of these issues is that tunnel ionization rates diverge from the true ones, which can be attributed to the assumption that ionization happens instantaneously. This can be fixed ad-hoc by introducing a pre-exponential scaling factor. This factor can be determined, by calculating the ionization rates in the limit of $\omega \rightarrow 0$ and then comparing these with experimental values or with other theories or TDSE simulations. This was already done for Keldysh ionization theory [29].

A significant issue is that the SFA is not gauge invariant. As explained in section 3.1, the Hamiltonian and eigenstates can be expressed in the Length or in the Velocity gauge. In any physical theory both gauges should predict the same observables, however after the many simplifications involved in the SFA this is not the case anymore. Experiments and comparisons to the TDSE have shown however that the length-gauge produces more accurate results, which is hence the reason why it used throughout this project and thesis. This is discussed in depth in [41], and well summarised in section 2.1.1 of MSci thesis [17].

Another issue is that the initial ground states $|\psi_g\rangle$ and the final Volkov states $|\psi_p^V\rangle$ are not completely orthogonal, unlike the true continuum $|\psi_p\rangle$ and ground states. This is due the Hilbert space spanned by

the Volkov plane-wave states (and associated Volkov time evolution operator $U_V(t, t')$) is only a subspace of the Hilbert space of the true continuum states. By 'forcefully' replacing $|\psi_p\rangle \xrightarrow{SFA} |\psi_p^V\rangle$, the system

now operates on an *incomplete* Hilbert space which breaks the orthogonality condition. This is problematic and leads to artifacts in the transition amplitudes expressions. It also means that the SFA is not invariant under spatial translations, which some refer to as the "Curse of the displaced atom" [42]. However this can be remedied by forcing orthogonality through substituting subspace propagators that ensure that each operator operates in the correct subspace. *over-complete* *correct* The NSDI SFA model uses this method and the details of such are outlined later. For further details and derivation of the remedying procedure see [42].

Bound states also do not evolve in time. The SFA neglects the external field before the ionization event, and as a consequence the electron stays in its ground states leaving out effects like excitation, core dynamics and field-induced distortions such as Stark Shifts [15].

However the biggest issue is the lack of dynamical behaviour due the constant drift momentum in the continuum. This means that more complicated processes like re-scattering can only be modelled by

not quite. This is related to the fact that $|\bar{p}'\rangle$ is already complete,

so that the SFA becomes overcomplete.

using higher order expansions as in explained in section 3.4. However this leaves out lots of other, more subtle dynamical behaviours like soft re-scattering. Even if the electron mostly oscillates ~~mostly~~ like a plane-wave, ~~it's~~ drift momentum is constantly affected by the residual Coulomb binding potential.

its

Apart from the over-completeness issue

it is a great accuracy

4 The Coulomb Quantum-Orbit Strong Field Approximation

The SFA so far has employed a plethora of different SFP scenarios such as NSDI, ATI and HHG where it is especially successful at modelling Quantum-mechanical features like interference [33]. However what all of these models have in common is the SFA's inherent flaw: the lack of the residual Coulomb binding potential in the continuum and the associated lack of dynamical behaviour. Given that the SFA is so commonly used that some even call it the "work-horse of SFP", it is not surprising that much effort has been made to develop extensions or models similar to the SFA which take this into account, resulting in plethora of different of Coulomb corrected theories. A comprehensive review of many of these theories can be found in [15]. This also the source of fig. 8 "family tree" of many of these methods.

4.1 Forward vs Backward Methods

Many above the above mentioned methods can be classified as "forward" or "backwards". The presence of the Coulomb potential in the continuum, means that an electron's initial drift momentum p_0 upon ionization and final drift momentum p_f at the detector are different. To practically plot momentum distributions and interference patterns, one must consider calculating the transition amplitudes M_{p_f} at some distribution (or 'box') of final momenta p_f .

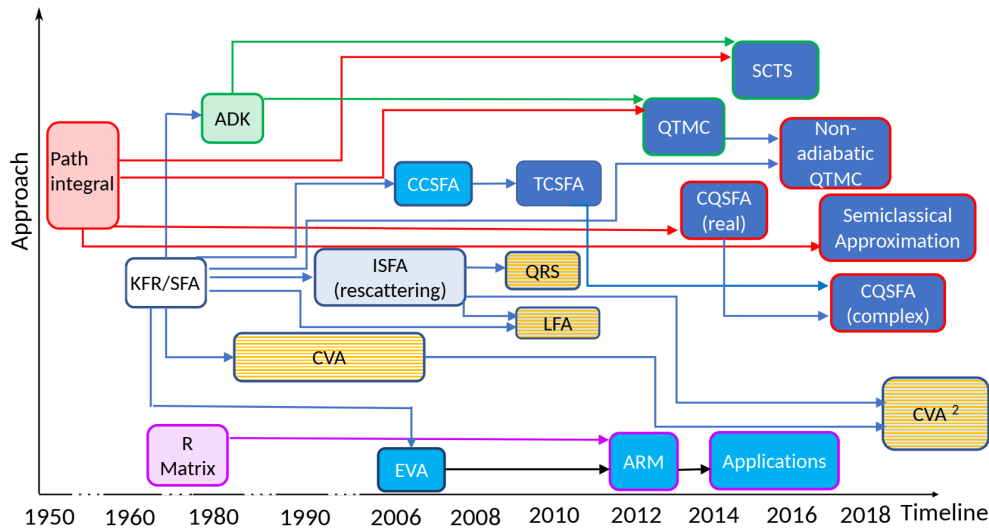


Figure 8: Family tree of different theoretical models that can be used in SFP. This diagram was included in order to show the diversity of different models available and their iterative and intertwined nature of their evolution. Copied from [15].

Forward methods solve the *direct* problem and start out with a uniform distribution of initial momentum points and use these as initial boundary conditions. The advantage is that the distribution of initial momentum is well known, generally making the problem easier to solve. The disadvantage is that this

usually results in an uneven distribution of final momentum points, meaning that interference patterns are easily washed out [15]. The non-conservative nature of the system means that predicting the final momentum of an quantum trajectory (or "orbit") given on the initial momentum is difficult. Low initial momenta may easily get mapped far outside the 'box' of final momentum points at which the transition amplitudes are to be calculated. An orbit also becomes unphysical if it encounters a singularity and diverges. As a result, a larger number of orbits is required to yield a sufficiently even distribution of final momentum points to be achieved. *(Good account but should specify how huge the number is: $10^8 - 10^9$ orbits)*

Backward methods solve the *inverse* problem by starting out with a uniform distribution of final momentum points, and use these as initial boundary condition. The initial momenta $\mathbf{p}_0$ of an orbit are therefore solved in terms of the final momentum. The advantage is that this leads to much clearer interference patterns. A significantly lower number of orbits is also required, since all orbits already have their final momentum in the desired range. Unfortunately solving the problem backwards is much more difficult, and requires some insight into how these orbits behave. How this is done for the CQSFA is explained in section 4.5.

I would talk here about ATI in general, as the direct / rescattered distinction becomes blurry

4.2 CQSFA Derivation

The CQSFA is a relatively modern Coulomb corrected SFA method. It retains a lot of the SFA's analytical nature, and in for a vanishing Coulomb potential collapses back to the SFA. *It has seen been extensively applied to High Order ATI where it handles the role of soft re-scattering, and predicts features such as the 'spider' and holographic side-lobes. For a comprehensive review of the CQSFA and how it compares to other theories see [15].*

The CQSFA aims to calculate the transition amplitude for a process for a process with initial ground state $|\psi_0\rangle$ and final continuum state $|\psi_{\mathbf{p}_f}(t)\rangle$ with drift momentum $\mathbf{p}_f$ at the detector at time t_f . For this we start out with equation (17) with updated notation $\tau \rightarrow t_0$ for the ionization time, which gives

$$M_{\mathbf{p}_f} = \lim_{t \rightarrow \infty} -i \int_{-\infty}^{t_f} dt_0 \langle \psi_{\mathbf{p}_f}(t_f) | U(t, t_0) V_{int}(t_0) | \psi_g(t_0) \rangle. \quad (35)$$

Further substituting the closure relationship $\mathbb{1} = \int d\tilde{\mathbf{p}}_0 |\tilde{\mathbf{p}}_0\rangle \langle \tilde{\mathbf{p}}_0|$ and writing the final continuum state as $|\psi_{\mathbf{p}_f}(t)\rangle = |\tilde{\mathbf{p}}_f\rangle$ gives

$$M_{\mathbf{p}_f} = \lim_{t_f \rightarrow \infty} -i \int_{-\infty}^{t_f} dt_0 \int d\tilde{\mathbf{p}}_0 \langle \tilde{\mathbf{p}}_f | U(t, t_0) | \tilde{\mathbf{p}}_0 \rangle \langle \tilde{\mathbf{p}}_0 | V_{int}(t_0) | \psi_g(t_0) \rangle \quad (36)$$

where $\tilde{\mathbf{p}}_0 = \mathbf{p}(t_0) + \mathbf{A}(t_0)$ and where $\mathbf{p}_0 = \mathbf{p}(t_0)$ can be considered the initial drift momentum at which the electrons enters the continuum upon ionization. In general $\tilde{\mathbf{p}}(\tau) = \mathbf{p}(\tau) + \mathbf{A}(\tau)$ still is the canonical momentum, the drift momentum $\mathbf{p}(\tau)$ is no longer considered to be constant due to the presence of Coulomb potential in the continuum.

The aim of the CQSFA is to express above equation in a form that is applicable to Saddle Point Method. This is easy for the second inner product term, as by substituting in $|\psi_g(t_0)\rangle = e^{iI_p t_0} |\psi_g\rangle$, we get

$$\langle \tilde{\mathbf{p}}_0 | V_{int}(t_0) | \psi_g(t_0) \rangle = V_{cg}(\tilde{\mathbf{p}}_0, t_0) e^{iI_p t_0} \quad (37)$$

which can be separated into a slow varying prefactor term $V_{cg}(\tilde{\mathbf{p}}_0, t_0) = \langle \tilde{\mathbf{p}}_0 | V_{int}(t_0) | \psi_g \rangle$ and a fast moving exponential term $e^{iI_p t_0}$. Note that the prefactor term is the same as the prefactor in the KFR amplitude (eqn. 20).

Getting the first inner product term into a form applicable to the SPA is a bit more tricky. By using a time-slicing method outlined in [43], in which the time evolution operator is sliced into $N + 1$ time order bits in the limit of $N \rightarrow \infty$ it is possible to re-express this as a path integral

$$\langle \tilde{\mathbf{p}}_f | U(t, t_0) | \tilde{\mathbf{p}}_0 \rangle = \int_{\tilde{\mathbf{p}}_0}^{\tilde{\mathbf{p}}_f(t)} \mathcal{D}' \tilde{\mathbf{p}} \int \frac{\mathcal{D}\mathbf{r}}{(2\pi)^3} e^{-i\mathcal{A}(\tilde{\mathbf{p}}, \mathbf{r}, t_0, t_f)}, \quad (38)$$

which contains an exponential term

$$\mathcal{A}(\tilde{\mathbf{p}}, \mathbf{r}, t_0, t_f) = \int_{t_0}^{t_f} d\tau \dot{\mathbf{p}}(\tau) \cdot \mathbf{r}(\tau) + \frac{1}{2} \tilde{\mathbf{p}}^2(\tau) + V_c(\mathbf{r}(\tau)). \quad (39)$$

Here $\mathbf{r}$ is the electron's position vector. Substituting the above expressions for the two bra-ket terms, gives transition amplitude expression (36) as

$$M_{\mathbf{p}_f} = \lim_{t_f \rightarrow \infty} -i \int_{-\infty}^{t_f} dt_0 \int d\tilde{\mathbf{p}}_0 \int_{\tilde{\mathbf{p}}_0}^{\tilde{\mathbf{p}}_f(t)} \mathcal{D}' \tilde{\mathbf{p}} \int \frac{\mathcal{D}\mathbf{r}}{(2\pi)^3} V_{cg}(\tilde{\mathbf{p}}_0, t_0) e^{iS(\tilde{\mathbf{p}}, \mathbf{r}, t_0, t_f)} \quad (40)$$

where all the exponential terms were collected into the new action

$$\begin{aligned} S(\tilde{\mathbf{p}}, \mathbf{r}, t_0, t_f) &= I_p t_0 - \mathcal{A}(\tilde{\mathbf{p}}, \mathbf{r}, t_0, t_f) \\ &= I_p t_0 - \int_{t_0}^{t_f} d\tau \dot{\mathbf{p}}(\tau) \cdot \mathbf{r}(\tau) + \frac{1}{2} \tilde{\mathbf{p}}^2(\tau) + V_c(\mathbf{r}(\tau)). \end{aligned} \quad (41)$$

The above expression now is in a desired form and ripe for the Saddle Point Approximation of the form of equations (24) and (30) to be applied to. This yields 3 saddle point equations. The first saddle point equation is

$$\frac{1}{2} \tilde{\mathbf{p}}_0^2(t_{0s}) + V(\mathbf{r}(t_{0s})) + I_p = 0 \quad (42)$$

and solves for the ionization times t_0 , and also reflects conservation of energy upon ionization. However due to the reasons explained in [43] the binding potential term $V(\mathbf{r}(t_{0s}))$ becomes negligible, therefore giving

$$\frac{1}{2} \tilde{\mathbf{p}}_0^2(t_{0s}) + I_p = 0 \quad (43)$$

which identical to the ionization time equation for the KFR amplitude (eqn. 25), and therefore shares the same general solution (eqn. 31).

The path integral SPA yields two saddle point equations

$$\dot{\mathbf{p}}(\tau) = -\nabla_r V_c(\mathbf{r}(\tau)) \quad (44)$$

$$\dot{\mathbf{r}}(\tau) = \mathbf{p}(\tau) + \mathbf{A}(\tau) \quad (45)$$

which solve for the drift momentum $\mathbf{p}$ and position $\mathbf{r}$ vector. These in fact are the Newton's classical equations of motion, but unfortunately do not have a known analytic solution requiring numerical methods. The boundary condition for the drift momentum must satisfy the same initial momentum $\mathbf{p}(t_0) = \mathbf{p}_0$ as in saddle point equation (43) and obviously the final momentum $\mathbf{p}(t_f) = \mathbf{p}_f$. The boundary condition

for the position vector technically is $\mathbf{r}(t_0) = 0$, but the situation is a bit more complicated, as explained in-depth later in the section. We can also rewrite saddle point equation (45) as an integral

$$\mathbf{r}(\tau) = \int_{t_0}^{\tau} \mathbf{p}(\tau) + \mathbf{A}(\tau) d\tau. \quad (46)$$

and by explicitly writing out saddle point equation (44), one finds that

$$\dot{\mathbf{p}} \cdot \mathbf{r} = -\mathbf{r} \cdot \nabla_r V_c(r) = V(r). \quad (47)$$

4.3 The Tunnel-Exit Approximation

To continue, the integral in the action (eqn. 41), which determines the electrons dynamics from time t_0 to t_f in the continuum, still needs to be evaluated. Doing this is not trivial because the solutions t_0 are complex meaning a specific integration contour through complex space is required. In theory, any contour that connects t_0 to t_f should be given the same value but due to the approximations used this is no longer the case [18].

Therefore the so-called "Tunnel-Exit Approximation" is employed to generate a suitable contour. This idea was first introduced in 2014 [44] and is employed by many other SFA like theories. The general idea is to split the contour into two distinct paths. The first path, referred as the "Tunneling path", is parallel to the imaginary axis and runs down from $t_0 = t_{0r} + it_{0i}$ to t_{0r} on the real axis. Here $t_{0r} = \Re t_0$ and $t_{0i} = \Im t_0$ are the real and imaginary parts of t_0 respectively. The second part is known as the "Propagation path" and simply runs along the real axis and connects t_{0r} to t_f . See fig. (9) for a visual demonstration.

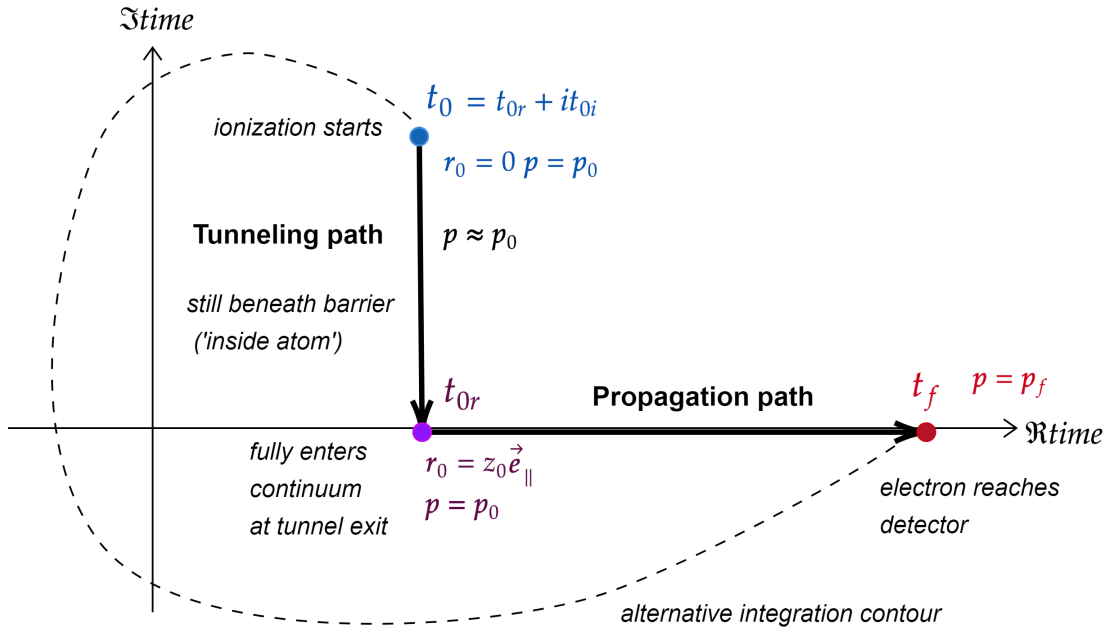


Figure 9: The integration contour used for the Tunnel exit approximation. The two thick black lines indicate the Tunneling and Propagation path that form the contour. The thin dashed line indicates a possible alternative integration contour. The original version of this diagram can be found in [44].

This specific choice of contour can be justified with physical intuition. As explained in section 3.8, the imaginary part of the ionization time is associated with the tunneling probability. A large imaginary

you can choose whatever contour you wish
if
the function is analytic.

Justification is OK but there is no need to

oh -
value
it.

part means that the electron is undergoing tunneling and can be treated as is if its position vector is located 'inside' the atom's potential barrier. Only for the zero imaginary part, the electron can assumed to be completely outside the barrier. That is also the reason why the final time t_f at which the electron reaches the detector is assumed to be purely real. Through splitting the contour into the two path defined above, we are effectively prescribing the electron to do all of its tunneling first before entering into and propagating about the continuum. Consequently the boundary condition for the position vector should be updated to reflect this interpretation. The position vector along the "tunneling path" is relabelled to $\mathbf{r}_0(\tau)$ and now specifically refers to the position inside the barrier. This still should still satisfy the original boundary condition $\mathbf{r}_0(t_{0r} + it_{0i}) = 0$ and start out at the origin. The position vector along the "propagation path" in the continuum still retains its original symbol of $\mathbf{r}(\tau)$. The electron no longer enters the continuum at the origin and instead enters it at some position outside the barrier on the z-axis known as the tunnel exit z_0 , at $\tau = t_{0r}$. This is where the two contours meet which serves as a boundary condition for two position vectors. Formally the tunneling exit is given by:

$$z_0 = \Re \mathbf{r}_0(t_{0r}) \cdot \vec{e}_z \quad (48)$$

Note that in the continuum $\mathbf{r}_0$ is always real and $\mathbf{r}_0(\tau)$ is complex and contains no real components other than the z-components. The tunnel-exit approximation is however not without its issues. Within the tunneling path, the Coulomb potential $V_c(\mathbf{r}) = \frac{c}{\sqrt{\mathbf{r} \cdot \mathbf{r}}}$ has a singularity at the origin and the multi-valued nature of $\sqrt{\mathbf{r} \cdot \mathbf{r}}$ in complex space of $\mathbf{r}$ means that branch cuts arise. These can be avoided by following the tunnel exit approximation contour described above until such a singularity or branch cut is found, in which case contour the carefully deviates around the singularity and returns back to the original path. Such a procedure is described in detail [45]. A more general study on the influence of integration contours on the transition amplitude can found in [46].

Here
it

This interpretation allows for further approximations on the action and saddle points. Along the **tunneling path** the electron should be sufficiently shielded so that its drift momentum remains constant, ie $\mathbf{p}(\tau) \approx \mathbf{p}_0$ which means that the $\dot{\mathbf{p}} \cdot \mathbf{r} \approx 0$ in the action (eqn. 41) vanishes. By writing the time along the tunneling path as $\tau = t_{0i} + i\tau_i$ (so that $d\tau = i d\tau_i$ where $t_{0i} \leq \tau_i \leq 0$) gives the tunneling action as:

ceases

$$S^{tun}(\tilde{\mathbf{p}}, \mathbf{r}, t_0, t_{0r}) = iI_p t_{0i} - i \int_{t_{0i}}^0 \frac{1}{2} \tilde{\mathbf{p}}_0^2(t_{0r} + i\tau_i) + V_c(\mathbf{r}_0(t_{0r} + i\tau_i)) d\tau_i. \quad (49)$$

to be analytic & the contour will matter

Note that momentum vector explicitly refers to $\tilde{\mathbf{p}}_0(\tau) = \mathbf{p}_0 + \mathbf{A}(\tau)$ and the position vector 'inside' the atom is hence given by

$$\mathbf{r}_0(t_{0r} + i\tau_i) = \int_{t_{0i}}^{\tau_i} \mathbf{p}_0 + \mathbf{A}(t_{0r} + i\tau'_i) d\tau' \quad (50)$$

Along the **propagation path**, identity (47) means that $\dot{\mathbf{p}} \cdot \mathbf{r} + V(r) = 2V(r)$ under which the action in the continuum becomes

$$S^{prop}(\tilde{\mathbf{p}}, \mathbf{r}, t_{0r}, t_f) = I_p t_{0r} - \int_{t_{0r}}^{t_f} \frac{1}{2} \tilde{\mathbf{p}}^2 A(\tau) + 2V_c(\mathbf{r}(\tau)) \quad (51)$$

where here a non-constant drift momentum means that explicitly $\tilde{\mathbf{p}}(\tau) = \mathbf{p}(\tau) + \mathbf{A}(\tau)$ and $\mathbf{r}(\tau) = \int_{t'}^{\tau} \mathbf{p}(\tau') + \mathbf{A}(\tau') d\tau'$.

4.4 CQSFA Quantum Orbits

As previously mentioned, the saddle point equations of the CQSFA (eqn. 43) and the SFA (eqn. 25) share the same general solution

$$t_{0s} = \frac{2\pi n}{\omega} \pm \frac{1}{\omega} \arccos \left(\frac{-p_{0\parallel} \mp i\sqrt{2I_p + p_{0\perp}^2}}{2\sqrt{U_p}} \right), \quad (52)$$

where $p_{0\parallel}$ and $p_{0\perp}$ are the parallel (ie z) and perpendicular components of the initial momentum vector $\mathbf{p}_0$. As explained in section (3.6), the SFA separates this solution into "long" and "short" orbits. For theories that use the Tunnel-exit approximation like the CQSFA, this logic can be extended to include two additional types of orbits. This definition is based on the relationship between the tunnel exit z_0 and the initial and final momentum of a trajectory.

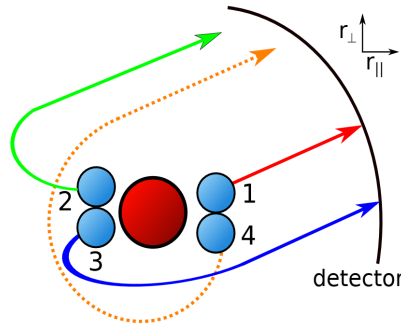


Figure 10: Visual demonstration of the physical interpretation of the 4 orbits. Original source: [18].

Orbit 1 (previously "short" SFA orbit) still represents the shortest possible trajectory to the detector. This means the electron enters the continuum on side of the atom closest to the detector and travels directly towards it. This will be the case of the final parallel momentum $p_{f\parallel}$ and the tunnel exit z_0 face the same direction, ie $p_{f\parallel}z_0 > 0$.

The SFA's "long" orbit now becomes **Orbit 2**. The electron enters the continuum at side opposite to the detector, and thus to reach the detector it must reverse its direction requiring that $z_0p_{f\parallel} < 0$.

Throughout orbit 1 and 2, the perpendicular momentum does not change direction, ie $p_{0\perp}p_{f\perp} > 0$. This condition ensures that influence of the Coulomb potential on the electron's trajectory remains minimal because a change in perpendicular momentum direction can only be facilitated by a sufficiently strong Coulomb force (by definition the electric field of the laser field does not have components in the perpendicular direction). Consequently the trajectories of Orbit 1 and 2 remain very similar to the SFA's original "long" and "short" orbits.

In Orbits 3 and 4 the introduction of the Coulomb potential represents a drastic departure in dynamical behaviour from the SFA, as here the influence of the Coulomb potential no longer remains weak (as $p_{0\perp}p_{f\perp} > 0$).

Like in orbit 2, electrons in **Orbit 3** spawn on opposite to the detector, but the Coulomb potential 'forcefully' accelerates the electron back towards the parent ion, hence $z_0p_{f\parallel} < 0$. This can result in ~~A~~ so called 'soft collisions', where the electron gets very close to the parent ion but without being reabsorbed

back into the ground state and instead is just deflected. This is especially important for understanding HATI.

In **Orbit 4** has its tunnel exit on the side closest to the detector, however rather than directly travelling towards the detector like in orbit 1, the Coulomb potential forces the electron to travel around the atom first, resulting in a 'very long' orbit. The tunnel exit and final parallel momentum still face the same direction however meaning that $z_0 p_{f\parallel} > 0$.

Orbit	$z_0 p_{f\parallel}$	$p_{0\perp} p_{f\perp}$
1	+	+
2	-	+
3	-	-
4	+	-

Table 1: Summary of definitions for 4 orbits of the CQSFA in terms of the tunnel exit z_0 , the final momentum along perpendicular z axis $p_{f\parallel}$ as well as $p_{0\perp}$ and $p_{f\perp}$ the initial and final momentum along perpendicular axis. Here + indicates > 0 and - means < 0 . This table is frequently employed CQSFA related papers such as [47].

The specific solutions for the ionization of the new orbit times are the same as (32) and (33) for the SFA. Here Orbit 1 still retains the solution

$$t_{0s} = \frac{1}{\omega} \arccos \left(\frac{-p_{0\parallel} - i\sqrt{2I_p + p_{0\perp}^2}}{2\sqrt{U_p}} \right), \quad (53)$$

and Orbit 2, 3 and 4 share the same solution

$$t_{0s} = \frac{2\pi}{\omega} - \frac{1}{\omega} \arccos \left(\frac{-p_{\parallel} + i\sqrt{2I_p + p_{\perp}^2}}{2\sqrt{U_p}} \right) \quad (54)$$

but differentiate themselves through different momenta [47]. Both of the above equations are for positive parallel momentum $p_{0\parallel} > 0$. The two equations swap place for negative momenta $p_{0\parallel} < 0$.

Following the completion of the path integral SPA yields the final CQSFA ATI transition amplitude as

$$M(\mathbf{p}_f) = -i \lim_{t_f \rightarrow \infty} \det \left[\frac{\partial \mathbf{p}_s(t_f)}{\partial \mathbf{p}_s(t_{0s})} \right]^{-1/2} \sqrt{\frac{2\pi i}{S''(\tilde{\mathbf{p}}_s, \mathbf{r}_s, t_{0s}, t_f)}} e^{iS(\tilde{\mathbf{p}}_s, \mathbf{r}_s, t_{0s}, t_f)}. \quad (55)$$

Here $S''(\tilde{\mathbf{p}}_s, \mathbf{r}_s, t_s, t_f)$ is the second derivative w.r.t ionization time t_0 , as can similarly be found for the standard SFA's KFR amplitude (eqn. 25). Here the $\det \left[\frac{\partial \mathbf{p}_s(t_f)}{\partial \mathbf{p}_s(t_{0s})} \right]^{-1/2}$ is the stability factor whose derivation is non-trivial and is explained here [43, 48]. In summary the original functional form of the Hessian as found in eqn. (30) simplifies down to vector operator $\partial \mathbf{p}_s(t_f)/\partial \mathbf{r}_s(t_s)$. Evaluating this expression is awkward though, and instead the Legendre transform is applied yielding $\partial \mathbf{p}_s(t_f)/\partial \mathbf{p}_s(t_s)$. This transform is legal and doesn't affect the action because the path $\mathbf{r}_s(t_s)$ starts at the origin. This factor is unique to Coulomb corrected SFA models and is important for explaining 'holographic sidelobes' in the momentum distributions in HATI [15].

justified (gerechtfertigt?)

4.5 CQSFA Orbit Solver

Calculating quantum trajectories for the CQSFA can only be done numerically, therefore an efficient C++ program was written by Dr. Andrew Maxwell to take care of this and also compute the interference. This program is sophisticated, featuring parallelization and intricate numerical methods for integration and interpolation. The solver considers a radial polar grid of *final* momentum points $(p_{f\parallel}, p_{f\perp})$ at which the transition amplitudes M_p are to be calculated. To solve this problem backwards and estimate the initial momentum, the solver exploits that in the limit of vanishing Coulomb potential (ie strength factor $C \rightarrow 0$), the CQSFA collapses to the standard SFA. This means it first solves the Continuum trajectories with the SFA's Volkov solutions (whose initial and final drift momenta are identical). Using this as an initial guess, an iterative numerical solver then solves the problem with the Coulomb potential's strength C increased to a small value δC . This procedure is repeated, each time increasing the C by δC and reusing the previous iteration as an initial guess, until the full strength of the Coulomb potential is reached. If a singularity or other diverging feature is encountered, the solver will dynamically create new points on the momentum grid to ensure that a solutions are always at a desired accuracy. As a result a structured grid of final momentum points is always ensured, however the distribution of initial momentum points also becomes more chaotic [?]. It is still not clear whether these solutions reached through this inverse methods are the same as if the problem is solved forward. *Good summary.*

5 The current NSDI SFA model

This section covers the basic derivation of the current NSDI model, and its predictions of the shape of the PMD for RESI. This derivation largely follows the same steps as the KFR amplitude in section 3.4 and thus a lot of the subtleties involved should already be clear. The only additional concepts are composite systems and the use of subspace propagators to force orthogonality as hinted in section 3.9.

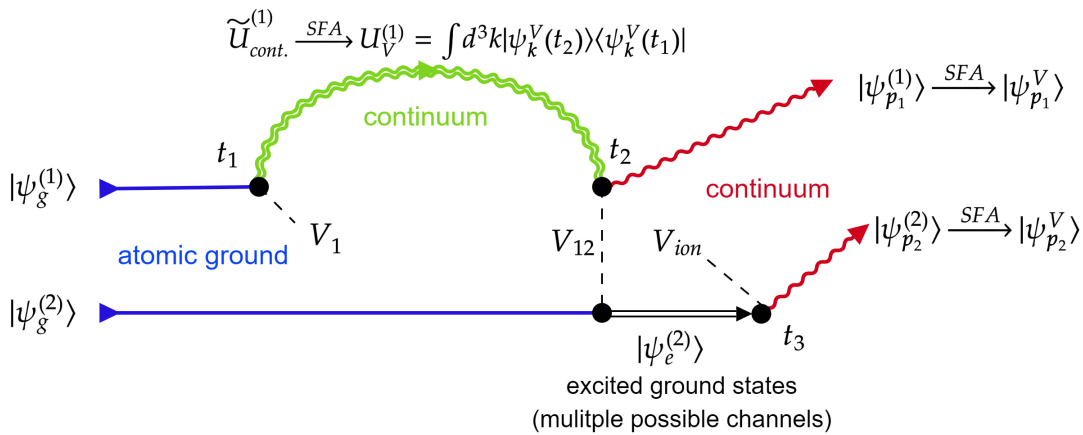


Figure 11: The Feynman diagram for the RESI process as used for the SFA model. Double lines, as used for the intermediate continuum transition of electron 1 (green) and excited state of electron 2, indicate that multiple values or pathways are considered in this diagram. Here $\tilde{U}_{cont}^{(1)}$ is the original exact time evolution operator for electron in its intermediate Coulomb transition.

The model for RESI starts out by considering the transition amplitude associated with the interaction shown in the Feynman like interaction diagram 11. The link between the Feynman diagram and transition amplitude of an interaction process was explained at the end of section 3.4. In this model both electron 1 and 2 remain in their atomic ground states ($|\psi_g^{(1)}\rangle$ and $|\psi_g^{(2)}\rangle$ with associated ionization potentials I_{1g} and I_{2g} respectively) until time t_1 , when electron 1 ionizes and enters the continuum. This event is modelled through the application of the Coulomb atomic binding potential operator labelled V_1 . Electron 2 will still remain in its ground state until time t_2 when electron 1 re-collides with parent ion. This collision causes electron 2 to be elevated into an excited ionic energy state $|\psi_g^{(1)}\rangle$ (with associated ionization potential I_{2e}) and electron 1 is scattered into a different continuum state of different momentum, and will eventually reach the detector. This is modelled through the electron-electron interaction operator labelled V_{12} . The second ionization event which propels electron 2 from its excited ionic state into the continuum happens at time t_3 , and is modelled through the ionic coulomb potential operator V_{ion} .

Following the application of the SFA, any continuum states and associated time evolution operators are replaced with Volkov plane wave states and Volkov time evolution operators, as indicated in fig. 11. Since the drift momentum in the Volkov states stays constant, both electrons are considered to directly enter the detector at time t_3 , as considering any later detection time $t > t_3$ will just add a Volkov phase term that does not affect dynamics of the system. The Volkov time evolution operator during time $t_1 \leq \tau \leq t_2$ for electron one's intermediate transition is written as

$$U_V^{(1)} = \int d^3k |\psi_{\mathbf{k}}^V(t_2)\rangle \langle \psi_{\mathbf{k}}^V(t_1)|, \quad (56)$$

where the integral $\int d^3k$ indicates that every possible intermediate drift momentum has to be considered, by applying the SPA later, this will effectively become a single intermediate drift momentum $\mathbf{k}$.

This process is integrated over every possible combination of event times that follow causality, ie $-\infty < t_1 \leq t_2 \leq t_3 < \infty$. Throughout this section the integration over the ionization time is summarised as

$$\int d^3t = \int_{-\infty}^{-\infty} dt_3 \int_{-\infty}^{t_2} dt_2 \int_{-\infty}^t dt_1 \quad (57)$$

which therefore allows the transition amplitude to be written as

$$M(\mathbf{p}_1, \mathbf{p}_2) = \int d^3t \langle \psi_{\mathbf{p}_1, \mathbf{p}_2}^V(t_3) | V_{ion}(t_3) \tilde{U}(t_3, t_2) V_{12}(t_2) U(t_2, t_1) V_1(t_1) | \psi_g(t_1) \rangle \quad (58)$$

The Hilbert space of the system is constructed as a compound of the sub-spaces of two electrons, which are appropriately labelled with the superscript. For instance the system's initial state is written as

$$|\psi_g(t_1)\rangle = |\psi_g^{(1)}(t_1)\rangle \otimes |\psi_g^{(2)}(t_1)\rangle. \quad (59)$$

Likewise the final state of the system at the detector is written

$$|\psi_{\mathbf{p}_1, \mathbf{p}_2}^V(t)\rangle = |\psi_{\mathbf{p}_1}^V(t)\rangle \otimes |\psi_{\mathbf{p}_2}^V(t)\rangle. \quad (60)$$

Similarly time evolution operators are also constructed as compounds, for instance the system's time evolution operator at time during time interval $t_1 \leq \tau \leq t_2$ is given by

$$U(t_2, t_1) = U_V^{(1)}(t_2, t_1) \otimes U_g^{(2)}(t_2, t_1) \quad (61)$$

where $U_V^{(1)}(t_2, t_1)$ is Volkov time evolution operator of electron 1 and $|\psi_g^{(2)}(t_1)\rangle$ is the time evolution operator of the ground state of electron 2. The time evolution operator during interval $t_2 \leq \tau \leq t$;

$$\tilde{U}(t_3, t_2) = U_V^{(1)}(t_3, t_2) \otimes U_e^{(2)}(t_3, t_2) \quad (62)$$

contains the time evolution operator for electron 2's excited state.

5.1 Subspace Projectors and Orthogonality

As explained in section 3.9, the Volkov continuum states are not orthogonal to the ground states, which can lead to artifacts in the interference patterns of the PMD's. The NSDI SFA model therefore uses the procedure developed in [42] to mitigate some of this effect. The methodology is quite simple: force orthogonality with subspace projectors so that each operator only operates within correct subspace.

A projector P_q of a specified state $|q\rangle$ (or set of states e.g. $|q_i\rangle$ for $i = 1, 2, 3, \dots$) is defined so that when applied to a general superposition of states e.g. $|\psi\rangle$, it will remove all states from $|\psi\rangle$ other than $|q\rangle$. In this example for the single state is explicitly defined as $P_q = |q\rangle\langle q|$ and for multiple states it would $P_q = \sum_i |q_i\rangle\langle q_i| = \sum_i P_{q_i}$. The sum over the projectors over all sub-spaces gives the identity, ie $\mathbb{1} = \sum_q P_q$. Another property that any projection operator P must obey is $P = P^2$. For more information see [49]. The subscript c indicates the "channel", or energy state that electron occupies. E.g. $c = g$ indicates the ground state and $c = e$ indicates an excited state. The propagator to a specific subspace such as the ground state would be $P_c = |\psi_g\rangle\langle\psi_g|$ and likewise for the excited state it would be $P_e = |\psi_e\rangle\langle\psi_e|$.

For the NSDI model, we want to clearly separate the ground state and excited states from the Volkov states. Hence the ground space projector is defined as $P_c^{(\mu)} = |\psi^{(\mu)}\rangle\langle\psi_g^{(\mu)}|$ and the excited state propagator is written as $P_e^{(\mu)} = |\psi_e^{(\mu)}\rangle\langle\psi_e^{(\mu)}|$ where $\mu = 1, 2$ indicates the subspace of which electron. The projector to momentum continuum subspace containing the Volkov states is simply written as:

$$P_c^{(\mu)} = \int d^3\tilde{p} |\tilde{p}\rangle\langle\tilde{p}|, \quad (63)$$

where the tilde signifies that this canonical and not drift momentum. These projectors also need to be defined as a compound for the subspace of each electron, which is defined by notation

$$P_{\alpha\beta} = P_\alpha^{(1)} \otimes P_\beta^{(2)}. \quad (64)$$

The next step of the procedure is to now 'pad' each operator with appropriate projectors. When applying V_1 at time t_1 , we initially consider the first electron to initially only be in its ground state and afterwards only be in the continuum state, and electron 2 to be in its ground state before and after. Enforcing this assumption with subspace projectors yields:

$$V_1 = P_{cg} V_1^{(1)} \otimes \mathbb{1}^{(2)} P_{gg}. \quad (65)$$

Likewise the other two operators should be written as

$$V_{ion} = P_{cc} \mathbb{1}^{(1)} \otimes V_{ion}^{(2)} P_{ce} \quad (66)$$

$$V_{12} = P_{ce} V_{12}^{(12)} P_{cg}. \quad (67)$$

Substituting all of the above into eqn. (58) and further simplification gives the transition amplitude as

$$M(\mathbf{p}_1, \mathbf{p}_2) = \sum_e \int d^3t \int d^3k V_{\mathbf{p}_2 e}(t_3) V_{\mathbf{p}_1 e, \mathbf{k}g}(t_2) V_{\mathbf{k}g}(t_1) e^{iS(\mathbf{k}, \mathbf{p}_1, \mathbf{p}_2, t_1, t_2, t_3)} \quad (68)$$

which is the in the desired form that is applicable to SPA. Here there are 3 prefactors associated with the first ionization event

$$V_{\mathbf{k}g}(t_1) = \langle \tilde{\mathbf{k}}(t_1) | V_1^{(1)}(t_1) | \psi_g(t_1) \rangle, \quad (69)$$

the recollision event

$$V_{\mathbf{p}_1 e, \mathbf{k}g}(t_2) = \left[\langle \tilde{\mathbf{p}}_1^{(1)}(t_2) | \otimes \langle \psi_e^{(2)}(t_2) | \right] V_{ion}^{(1,2)}(t_2) \left[| \tilde{\mathbf{k}}^{(1)}(t_2) \rangle \otimes | \psi_g^{(2)}(t_2) \rangle \right], \quad (70)$$

and the final excitation event

$$V_{\mathbf{p}_2 e}(t_3) = \langle \tilde{\mathbf{p}}_2(t_3) | V_{ion}^{(2)}(t_3) | \psi_e(t_3) \rangle. \quad (71)$$

Calculating the explicit forms of these prefactors is complicated as it involves solving hydrogenic wave state functions. See [50] and appendix C of PhD thesis [30] for more informatino. However for the Frankenstein model, the prefactors are ignored anyway, as we are mainly interested in the dynamics of the exponential factors. The action is written as:

$\leftarrow$ *Space*

$$S(\mathbf{k}, \mathbf{p}_1, \mathbf{p}_2, t_1, t_2, t_3) = S_1(\mathbf{k}, \mathbf{p}_1, t_1, t_2) + S_2(\mathbf{p}_2, t_3) \quad (72)$$

which is separated into the action of electron 1

$$S_1(\mathbf{k}, \mathbf{p}_1, t_1, t_2) = I_{1g}t_1 + (I_{2g} - I_{2e})t_2 - \int_{-\infty}^{t_1} d\tau \frac{1}{2} [\mathbf{p}_1 + \mathbf{A}(\tau)]^2 \int_{t_1}^{t_2} d\tau \frac{1}{2} [\mathbf{k} + \mathbf{A}(\tau)]^2 \quad (73)$$

and electron 2

$$S_2(\mathbf{p}_2, t_3) = I_{2e}t_3 - \int_{-\infty}^{t_3} d\tau \frac{1}{2} [\mathbf{p}_2 + \mathbf{A}(\tau)]^2. \quad (74)$$

The goal is to now apply the multivariable SPA, as dictated by equation (27). These should be applied to approximate the integrals over the 3 ionization times (t_1, t_2, t_3) and the three components of the intermediate momentum $\mathbf{k}$. The 3 saddle point equations for the components of $\mathbf{k}$ can be summarised as a single vector based equation

$$\mathbf{k} = -\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \mathbf{A}(\tau) d\tau. \quad (75)$$

This can be interpreted as the average change in momentum brought by the field during the first ionization and collision event, and thus is purely a function of the ionization times (t_1, t_2) . The saddle point equations for the 3 ionization times are

$$[\mathbf{k} + \mathbf{A}(t_{1s})]^2 = -2E_{1g} \quad (76)$$

$$\tilde{\mathbf{p}}_1^2(t_{2s}) = [\mathbf{k} + \mathbf{A}(t_{2s})]^2 - 2(E_{2e} - E_{2g}) \quad (77)$$

$$\tilde{\mathbf{p}}_2^2(t_{3s}) = -2E_{2e} \quad (78)$$

where $\tilde{\mathbf{p}}_\mu(\tau) = \mathbf{p}_\mu + \mathbf{A}(\tau)$. However according to eqn. (75), the two components of k in the perpendicular x-y plane will be zero. This is problematic because then the determinant of the 6x6 Hessian Matrix will be zero, which leads the expression to fail. A simple work around is to first apply the SPA only to the 3

components of $\mathbf{k}$, and then afterwards to the 3 ionization times. The first step leads to the same saddle point equation for $\mathbf{k}$ as before (eqn. 75), and the determinant of the 3x3 Hessian matrix boils down to

$$\begin{vmatrix} \frac{\partial^2 S}{\partial k_1^2} & \frac{\partial^2 S}{\partial k_1 \partial k_2} & \frac{\partial^2 S}{\partial k_1 \partial k_3} \\ \frac{\partial^2 S}{\partial k_2 \partial k_1} & \frac{\partial^2 S}{\partial k_2^2} & \frac{\partial^2 S}{\partial k_2 \partial k_3} \\ \frac{\partial^2 S}{\partial k_3 \partial k_1} & \frac{\partial^2 S}{\partial k_3 \partial k_2} & \frac{\partial^2 S}{\partial k_3^2} \end{vmatrix} = \begin{vmatrix} t_1 - t_2 & 0 & 0 \\ 0 & t_1 - t_2 & 0 \\ 0 & 0 & t_1 - t_2 \end{vmatrix} = (t_1 - t_2)^3. \quad (79)$$

The following application of the SPA to the 3 ionization times also leads to the same saddle point equations as before (eqns. 76- 78). As a result the intermediate momentum $\mathbf{k}$ is now purely a function of the ionization saddle point times t_{1s} and t_{2s} , ie $\mathbf{k} \rightarrow \mathbf{k}_s = \mathbf{k}(t_{1s}, t_{2s})$. The transition amplitude finally reads

$$\begin{aligned} M(\mathbf{p}_1, \mathbf{p}_2) &= \sum_s A_s e^{iS(\mathbf{k}_s, \mathbf{p}_1, \mathbf{p}_2, t_{1s}, t_{2s}, t_{3s})} \\ &= \sum_s \sum_e (2\pi)^3 \frac{V_{\mathbf{p}_2 e}(t_{3s}) V_{\mathbf{p}_1 e, \mathbf{k}g}(t_{2s}) V_{\mathbf{k}g}(t_{1s})}{\sqrt{(t_{1s} - t_{2s})^3 \det S''(\mathbf{k}_s, \mathbf{p}_1, \mathbf{p}_2, t_{1s}, t_{2s}, t_{3s})}} e^{iS(\mathbf{k}_s, \mathbf{p}_1, \mathbf{p}_2, t_{1s}, t_{2s}, t_{3s})}, \end{aligned} \quad (80)$$

where all prefactors can be collected into a single prefactor term A_s . The summation over the excitation channels e is commonly left out in this expression, as will be done here from now on.

5.2 Solving NSDI orbits

The practical implementation of the NSDI model is a bit more complicated than for the direct case, as the saddle point equations (76) and (77) for time t_1 and t_2 are coupled system of equations that unfortunately has no obvious analytic solution requiring numerical methods. To handle these numerical implementation a C++ program was developed by the research team of Prof. Faria. This program also takes care of all possible interference. This creates a rectangular 'box' of discrete momentum points (columns containing constant parallel momenta $p_{\parallel}$ and rows containing constant perpendicular momenta $p_{\perp}$, see section. 6.4), along which the Transition amplitude and Partial Momentum Distributions are calculated.

The program uses iterative numerical solvers which take two initial guess value, and then iterative approach the solutions for t_{1s} and t_{2s} . To generate initial guesses, the Simple Man's Model is used, according to which ionization happens around the peaks (at time $t_{peak} = \frac{\pi}{4\omega}$) and recollision around the crossing (time $t_{cross} = \frac{\pi}{2\omega}$) of the electric field $E(t)$ as explained in section 3.7. To emulate the SFA's Orbit 1 (short) and 2 (long) that are found in analytic solution, two sets of solution trajectories (orbits) are also considered numerically. In the long orbit, the electron ionizes before and returns later than the electron in the short orbit, as visually explained in 12. Therefore the first initial guesses for the ionization t_1 and return times t_2 of Orbit 1 (short) are $t_{peak} + \epsilon_{1a}$ and $t_{cross} - \epsilon_{1b}$ respectively and likewise $t_{peak} + \epsilon_{2a}$ and $t_{cross} - \epsilon_{2b}$ for Orbit 2 (long). Here $0 < \epsilon_{1a} < \epsilon_{2a}$ and $0 < \epsilon_{1b} < \epsilon_{2b}$ are small values that are adjusted 'by hand' to give relevant predictions. The orbit solver uses this guess value a single point on the momentum grid. It then takes solved value of the initial point and uses them as initial guesses for the adjacent point. The idea is that points with similar momenta should also have similar ionization times. The program repeats this procedure by moving along columns points with increasing perpendicular momentum $p_{\perp}$ and constant parallel momentum $p_{\parallel}$, until it hits the edge of the momenta grid, in which case it jumps to the next column (ie a next available jump in $p_{\parallel}$), and solves the point downwards with decreasing parallel

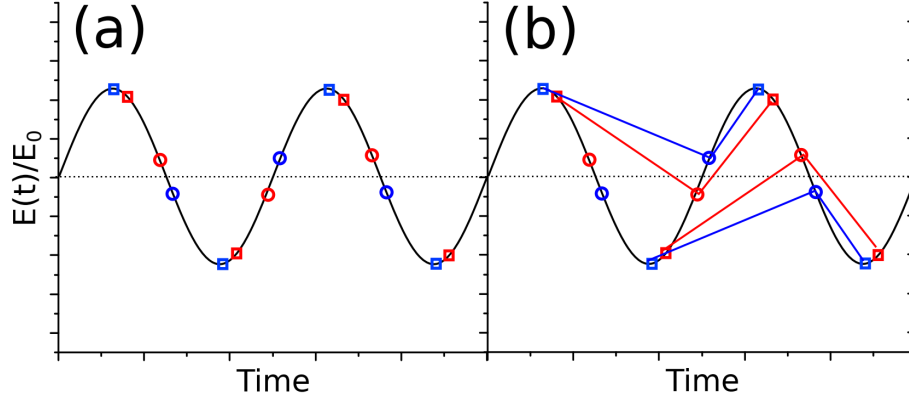


Figure 12: Possible long and short orbits for the NSDI SFA. Panel a) shows possible ionization (squares) and collision events (circles). Panel b) shows the possible long (blue) and sort orbits (red) that can be formed from them. The first line indicates electron one's intermediate transition, whereas the end of the second line indicates the ionization of electron 2. Copied from [18].

momenta $p_{\perp}$). This procedure is repeated until the ionization times for t_2 and t_3 are solved for the entire grid.

Ideally the first initial point for the guess times should lie in a region known as the **classical region**. This is defined as the region in momentum space whose saddle ionization time solutions have a almost vanishing imaginary part. Hence the ionization process in this region is (or at least almost is) classically allowed, and the Simple Man's model should be the most appropriate here. However to account that the first initial guess point might now fully be inside the classical region, small imaginary parts to the guess time are considered which are also 'adjusted by hand' and hard coded. Within the classical region, the real part of the action actions of orbit 1 and 2 are very similar, and tend intersect at locations not far beyond the classical region. The region located between the classical region and this intersection is known the **Stokes transition**. Beyond the ζ Stokes transition, the transition amplitudes (according the standard SPA described above) for at least one of the orbits tends to diverge very quickly (the other orbit usually diverges). The divergence is problematic as an infinitely increasing probability for a process is unphysical. The reason why the standard SPA fails is because imaginary parts of the saddle point times are located very close together in momentum space, meaning that the Gaussian approximation curves overlap as hinted in sec. 3.5. For more information see [18, 30].

5.3 The Uniform SPA

This problem can be fixed by using the "Uniform Saddle Point Approximation", whose details are explained in [51]. It works similar to the standard SPA however it considers higher expansions of the action S in the exponential, to account for two closely coalescing saddle points. The solution to the integral of this new approximate form is analytic, but won't be a Gaussian anymore and involves Bessel functions of type K and J. The Uniform SPA (USPA) only needs to be applied to a pair of long and short orbits for electron 1. The solution is also piece-wise, with two different expressions existing for within and outside Stokes transition region. For a pair of long and short orbits (denoted by i and j) the transition amplitude

under the uniform SPA becomes:

$$M_i + M_j = A_i e^{S_i} + A_j e^{S_j} \xrightarrow{USPA} M_{i+j} \quad (81)$$

where for amplitude inside and outside the Stokes Transition region is given by

$$M_{i+j}^{inside} = \sqrt{\frac{2\pi\Delta S}{3}} e^{i\bar{S} + i\frac{\pi}{4}} \bar{A} [J_{1/3}(\Delta S) + J_{-1/3}(\Delta S)] + \Delta A [J_{2/3}(\Delta S) - J_{-2/3}(\Delta S)] \quad (82)$$

$$M_{i+j}^{outside} = \sqrt{\frac{2\pi\Delta S}{\pi}} e^{i\bar{S}} [\bar{A} K_{1/3}(-i\Delta S) + i\Delta A K_{2/3}(-i\Delta S)] \quad (83)$$

respectively. Here J_α is the Bessel function of the first kind and K_α modified Bessel functions of the second kind. All other terms are defined by

$$\begin{aligned} \Delta S &= (S_i - S_j)/2, & \bar{S} &= (S_i + S_j)/2 \\ \Delta A &= (A_i - A_j)/2, & \bar{A} &= (A_i + A_j)/2 \end{aligned} \quad (84)$$

5.4 Momentum Constraints and Interference

It possible to predict the basic shapes and features of the **Partial Momentum Distributions (PMDs)** of RESI analytically. The idea is to apply well known momentum cut-off laws predict by the Simple Man's Model to saddle point equations (77-78) to constrain the possible values for electrons momentum $\mathbf{p}_1$ and $\mathbf{p}_2$. Based on these constraints, it is possible to predict the regions on the PMDs where non-zero transition amplitude and also where quantum interference could happen. Throughout this section, it is useful to remind ourselves of the explicit form of electric fields' transition amplitude (for a real time τ)

$$A(\tau) = \frac{E_0}{\omega} \cos(\omega\tau) = 2\sqrt{U_p} \cos(\omega\tau). \quad (85)$$

The first step is to estimate the limits of parallel momentum $p_{2\parallel}$ of **electron 2**. Looking at the Feynman like diagram 11 and the transition amplitude equation (80) it should be easy to see that the transition of electron 2 from its excited state to the continuum is independent of electron 1 and is very similar to that of (direct) ATI. Therefore electron 2 is subject to the same momentum cut-off law for direct ATI, which predicts the

$$-2\sqrt{U_p} < p_{2\parallel} < 2\sqrt{U_p}. \quad (86)$$

This result is predicted by classical ionization theory and is also experimentally verified [2]. However it also becomes apparent from saddle point equation (78) which can be rearranged to

$$[\mathbf{p}_2 + \mathbf{A}(t_{3s})]^2 = -2E_{2e} \iff p_{2\parallel} = i\sqrt{p_{2\perp}^2 + 2I_{2e}} + A(t_{3s}) \quad (87)$$

and that $p_{2\perp}$ must be real so that therefore $-2\sqrt{U_p} \leq A(t_{3s}) \leq 2\sqrt{U_p}$. Note the first term is imaginary and is cancelled out by the imaginary part of $A(t_{3s})$.

Doing the same for **electron 1** is more complicated, as we also need to predict the values of in the canonical intermediate momentum $\tilde{\mathbf{k}}(t_{2s})$. Here again we can also use another well known limit commonly referred to as the "10 Up cutoff" predicts that electrons re-scatter with a kinetic energy of about

$$\frac{1}{2} \tilde{\mathbf{k}}^2(t_2) \approx 10U_p. \quad (88)$$

as derived here [52]. We can also derive this explicitly from the equations used in the previous section

$$\tilde{k}(t_2) = k_s(t_2, t_1) + A(t_2) = \frac{-1}{t_2 - t_1} \int_{t_1}^{t_2} A(\tau) d\tau + A(t_2) = \left(\frac{-\sin(\omega t_2) + \sin(\omega t_1)}{\omega t_2 - \omega t_1} + \cos(\omega t_2) \right) 2\sqrt{U_p}. \quad (89)$$

Plugging in the values of $\omega t_1 = \pi/2$ (ionization at field maximum) and $\omega t_2 = \pi$ (return at field crossing) as predicted by simple mans model will yield the limit $k \sim 3.27\sqrt{U_p}$. Please note that is slightly higher than what predicted by the "10 Up" law ($k \sim 3.17\sqrt{U_p}$). In reality ionization time and return times are not exactly at the field maximum and crossing respectively. The "10 Up" law uses conditions on based on the position vectors further constrain to further constrain the ionization time. Using this limit, the saddle point equations for electron 1

$$[\mathbf{p}_1 + \mathbf{A}(t_{2s})]^2 = \tilde{\mathbf{k}}^2(t_{2s}) - 2(I_{2e} - I_{2g}), \quad (90)$$

which was previously eqn. (77), can be rearranged to

$$(p_{1\parallel} \pm 2\sqrt{U_p})^2 + p_{1\perp}^2 \leq 2 \times 10U_p - 2(I_{2e} - I_{2g}) = 2\Delta E \quad (91)$$

which is the equation of (two) circles centred around $\pm 2\sqrt{U_p}$ and radii $\sqrt{2\Delta E}$. Here the term ΔE is the change in kinetic energy electron 1 experiences through the inelastic re-scattering, and its definition should be obvious from the above equation. The radii

$$\sqrt{2\Delta E} = \sqrt{20 - \frac{2}{U_p}(I_{2e} - I_{2g})} \times \sqrt{U_p} \quad (92)$$

of these two circles increases with the ponderomotive energy U_p , and can theoretically approach a maximum $\sim 2\sqrt{5} \times \sqrt{U_p}$ for very large U_p as can be seen above. However at very large values the "10 Up" cut-off might become unreliable. It also decreases slightly with increasing ionization potential of the excited state I_{2e} but this effect becomes negligible at larger U_p . Note that for $I_{2e} - I_{2g} < 3.17U_p$ the cut-off law also becomes unreliable due there being not enough energy to promote electron 2 into a physically allowed excited state [30]. Considering zero value of $p_{\perp}$ gives the maximum range of parallel perpendicular momentum, which is

$$\pm 2\sqrt{U_p} - \sqrt{2\Delta E} < p_{1\parallel} < \pm 2\sqrt{U_p} + \sqrt{2\Delta E}. \quad (93)$$

Note that because the PMD's integrate over the perpendicular momentum $p_{\perp}$ for each value of $p_{\parallel}$, the largest values of the PMDs should be found at $p_{1\parallel} = \pm 2\sqrt{U_p}$ (centre of the circles), as this value of $p_{1\parallel}$ has the highest range of possible $p_{1\perp}$ that an electron can occupy at detector. Combining these limits for $p_{1\parallel}$ and $p_{2\parallel}$ allows one to draw the basic shape of the PMD which turns will be two 'boxes' of width $2\sqrt{2\Delta E}$ and height $4\sqrt{U_p}$ centred at $(\pm 2\sqrt{U_p}, 0)$ as indicated in red in fig. 13. However in the detector electron 1 and 2 are indistinguishable, which therefore means need to take into account the addition of exchange in momentum $\mathbf{p}_1 \rightleftharpoons \mathbf{p}_2$ of electron 1 and 2. This effectively means reflecting the PMD shape across the $p_{2\parallel} = p_{1\parallel}$ axis, leading to two extra 'boxes', indicates in blue in the figure. At low U_p these boxes these boxes don't overlap meaningfully, but this quickly changes as U_p increases as then the 4 boxes significantly overlap, forming a shape commonly referred to as the "cross. In these 'overlap' regions, lots of interference happens, making these regions the most interesting. With appropriately sized values of

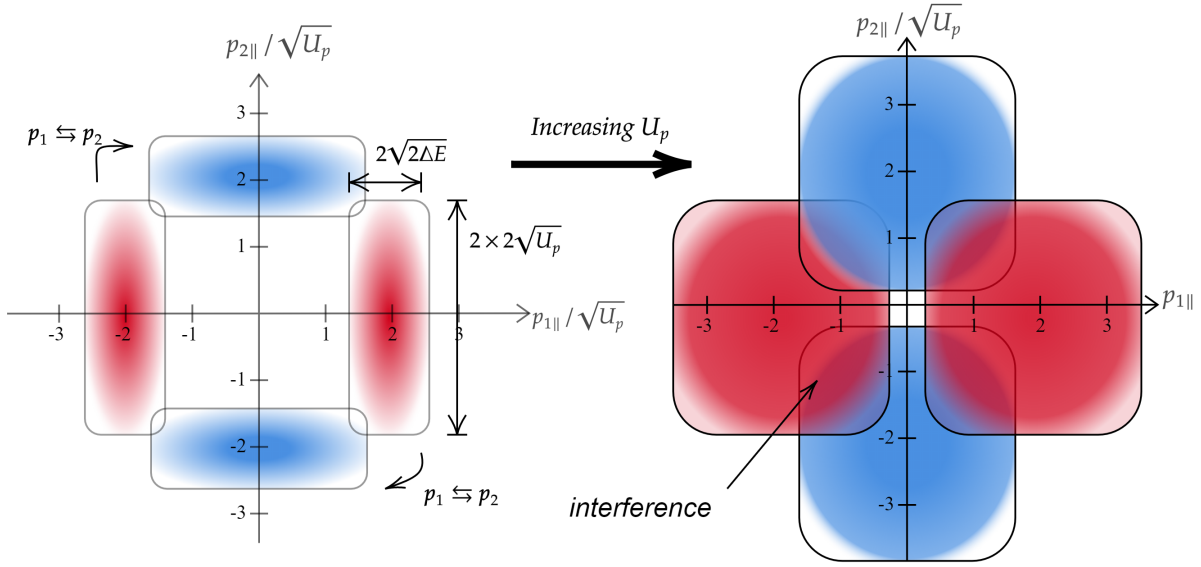


Figure 13: The shapes of the Partial Momentum Distribution (PMD) as predicted by the momentum constraints. The two red boxes indicate regions originally predicted, where blue indicates that exist only after considering exchange in momentum $\mathbf{p}_1 \rightleftharpoons \mathbf{p}_2$. With larger ponderomotive energy U_p the four boxes overlap significantly forming the cross. Regions of overlap is where most of the interference is expected. All momenta in units of $\sqrt{U_p}$.

U_p , the 'cross' experiences a small hole or momentum suppression as indicated in fig. 13. But for most values large of U_p which still allows for RESI, the 'hole' disappears as all 4 boxes overlap all across the centre.

The shapes however predicted by the momentum constraints are only approximate. For instance they only predict *where* interference *could* potentially happen, but not *how* it would look like, which can only be achieved numerically by fully considering the entire transition amplitude properly. The Orbit solving program however still uses these predictions as it calculates the interference independently for each of those 4 boxes. This allows one to dissect how the amplitude in each individual box affects the cross patterns. Note that different 'events' (ie different set ionization time solutions, that are usually separated by half a cycle) may favour to be in different boxes. Hence this interference between those 4 boxes is also known as "event" interference and can affect the symmetry of the PMD. Naturally, interference the PMDs predicted with SFA RESI have two axes of symmetry, leading to so-called "fourfold symmetry". Through considering the interference between different events, this fourfold symmetry can be broken, leading to possible variety of different shapes as explored here [20, 26, 53].

Not quite - there are approximate expressions for event interf.

Another aspect that massively influences the shapes in the PMDs is 'channel interference'. In the RESI mechanism, the second electron has a choice of different excited state configurations it may occupy, each of which has a Ionization potential. Transition laws provide some restrictions on excited states, meaning that only some allow for a transition (or "channel") into the continuum. By considering the superposition of multiple channels simultaneously (manifested through sum $\sum_e$ in equation 80) leads to a lot of quantum interference and consequently a myriad of possible shapes in the PMDs, as illustrated in fig. 14. The field parameters also influence the probability of a channel; for instance s states prefer short pulses whereas p and d prefer longer ones [26]. Consequently it should be possible to directly control the

quantum-interference and excited state geometry with the appropriate laser configuration which could be of great utility for practical applications. This aspect and analytic conditions for channel and event interference have been explored in [20, 26].

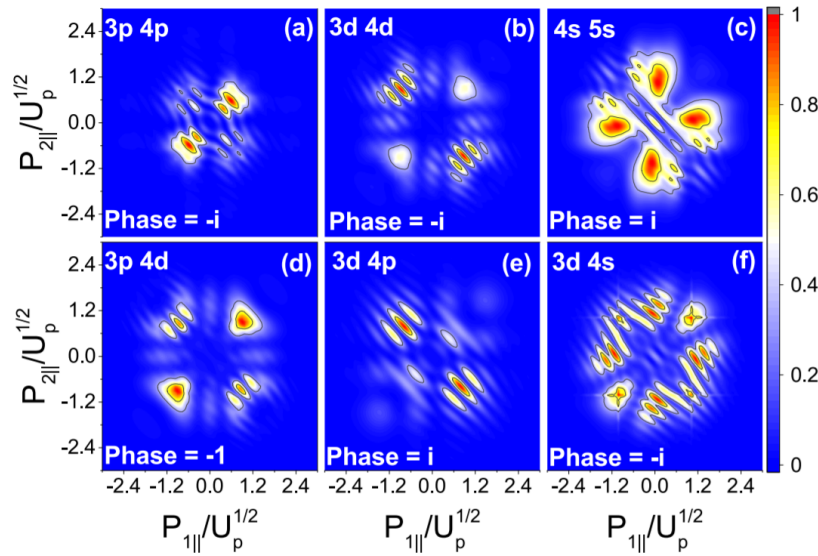


Figure 14: Partial Momentum Distributions displaying the interference between two different possible excitation channels (listed in top left of each diagram) in Argon Ar+ ions. The phase difference the channels is listed at the bottom. Source: [20].

6 The New 'Frankenstein' Model

6.1 The need for a new model

The current NSDI SFA model unfortunately inherits the fundamental issues of the SFA as outlined in section 3.9. The most notable of which is the lack of the Coulomb potential in the continuum, which leads to a reduction in dynamical behaviour and thus potentially to features of PMD being lost. As shown in the previous section, there are a lot of subtleties that can can drastically affect quantum interference, hence a proper treatment of the Coulomb potential in the continuum could further affect this. So far there has only been some effort to fully take the Coulomb potential into account for NSDI RESI, with best example being the Quantitative Rescattering Theory (QRS) [54], which however is not a full SFA based model as it also incorporates other models ad-hoc.

There ~~is~~ ^{are} also some experimental developments that challenge the current RESI model. For example both [55] and [56] claim results which cannot be explained by RESI. However the second paper's claims relies the predictions made by the standard SFA RESI model. Therefore to really answer the question of whether there is a fundamentally new mechanism, requires more accurate predictions for RESI mechanism. A step in the right direction would be to give RESI SFA model a full Coulomb treatment.

The CQSFA is an excellent candidate to improve this, as it largely retains the SFA's analytical nature and ~~and~~ collapses to the standard SFA for a vanishing Coulomb potential. This is important because it means that the fundamental predictions of the current RESI model should still be remain relevantt in

— Quite long intro
to set up the problem.
Demonstrates a great
level of understanding
and attention to detail

relevant

the new model.

6.2 Aims and Objectives

The general aim of this project was to extend the RESI SFA model by incorporating the CQSFA, and investigate the fundamental changes. The specific objectives were to calculate the Partial Momentum Distribution for a variety of different field and ionization potential parameters, and identify any new features. This was to be followed with closer investigation into these new features and the development of new analytic momentum constraints. This was intended as a pilot study, and as such a variety of simplifications were expected to be made. The most significant of these were neglecting the prefactors and only implementing the model for electron 2.

Throughout this project, I have worked closely together with Gergely Eöry who also did this project for his MSci thesis. I have focused mostly on developing the formal derivation (section 6.3). The analysis in section 7.2 and 7.2 is my own work. He has mostly focused mostly on coding and numerical implementation. However there was some overlap between both areas, for instance I have prototyped some of the features for the numerical implementations in Mathematica but these were not used in the end. Individual contributions are clearly indicated.

6.3 Theory

In this model electron 2's transition from the excited state to the detector in the continuum is calculated using the CQSFA and leave the rest of interaction is left untouched. So why just give electron 2 a proper CQSFA treatment and not electron 1? Firstly, implementing the CQSFA is non-trivial due to the more difficult implementation of the numerical computation of orbits and interference. So it makes sense to only implement it where the largest impact is expected. As shown in section 5.4, small subtleties in the excited state geometry and dynamics of electron 2 massively affect interference and shapes of the PMDs, at least when compared to electron 1. We anticipated that the inclusion of the Coulomb potential for the electron 2 would have the largest impact on the overall interaction. Implementing the CQSFA on electron 2 is also easier compared to electron 1, because the transition from the excited state to the continuum detector appears to be very similar to that of the ATI process. The main difference being that this transition begins in an excited rather than ground state, which effectively only constitutes to a shift in the value of the ionization potential $I_g \rightarrow I_e$. This means the existing Orbit Solver code for the CQSFA ATI, including its intricate regularization and divergence avoidance features, can mostly be reused with little alteration required. A lot of these features would have to be re-coded for electron 1. Because this new model is part SFA, part CQSFA we have temporarily named it the "FrankenStein". The formal derivation of this RESI FrankenStein transition amplitude becomes clear when looking at Feynman diagram 15. The transition amplitude expression remains identical to that of the RESI SFA up until the ionization event of electron 2 at time t_3 . Rather than both electrons straight entering the detectors after ionization t_3 , electron 2 travels through the continuum first during which its drift momentum will be affected by the interaction residual coulomb binding potential. Electron 2 now enters the detector at time t_f (in the limit of $t_f \rightarrow \infty$). Consequently an additional time evolution operator has

discussed, maybe?

Formally what you have

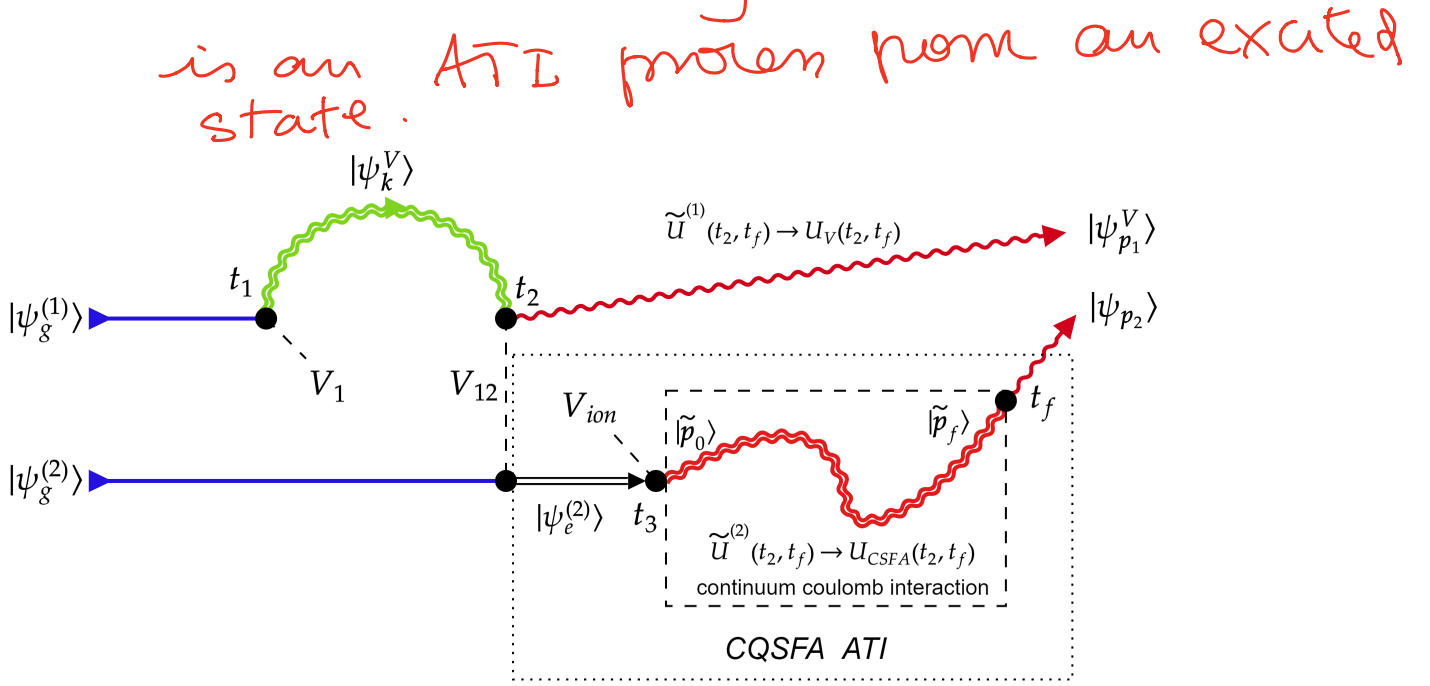


Figure 15: Feynman diagram of the new Frankenstein Model. All interactions and expressions remain the same for the NSDI SFA model, except for those in the dashed box, which indicates the interaction the is modelled by the CQSFA ATI. Double lines indicate that multiple path ways are considered and integrated over. The straight line for electron 1 indicates the constant drift momentum of the Volkov states. Likewise the continuum path for electron 2 is shifting to indicate that the drift momentum is affected by the coulomb potential in the continuum. Note that $\mathbf{p}_0$ and $\mathbf{p}_f = \mathbf{p}_2$ are the initial and final drift momentum of electron two's continuum transition.

to be added in order to model interactions during $t_3 < \tau < t_f$:

$$\tilde{U}(\tau, t_3) = U_V^{(1)}(\tau, t_3) \otimes U_{CQSFA}^{(2)}(\tau, t_3). \quad (94)$$

Here $U_V^{(1)}(\tau, t_3)$ is the Volkov time evolution operator for electron 1 and $U_{CQSFA}^{(2)}(\tau, t_3)$ is the continuum time evolution operator as modelled by the CQSFA which takes the Coulomb potential into account and remains identical to eqn. (36). The transition amplitude is written as

$$M(\mathbf{p}_1, \mathbf{p}_2) = \lim_{t_f \rightarrow \infty} \int d^3t \langle \psi_{\mathbf{p}_1}, \mathbf{p}_2(t_f) | \tilde{U}(t_f, t_3) V_{ion}(t_3) \tilde{U}(t_3, t_2) V_{12}(t_2) U(t_2, t_1) V_1(t_1) | \psi_g(t_1) \rangle, \quad (95)$$

where the sum over excited states has been omitted. Following the same procedures as in the NSDI SFA and CQSFA for electron 1 and 2 respectively, the transition amplitude can be expressed in terms of prefactors and exponential actions terms

$$M(\mathbf{p}_1, \mathbf{p}_2) = -i \int d^3t \int d^3k V_{\mathbf{p}_0 e}(t_3) V_{\mathbf{p}_1 e, \mathbf{k}g}(t_2) V_{\mathbf{k}g}(t_1) e^{iS(\mathbf{k}, \mathbf{p}_1, \mathbf{p}_0^{(2)}, \mathbf{p}_f^{(2)}, t_1, t_2, t_3)}, \quad (96)$$

which contain the same prefactors as the NSDI RESI model (eqns. 69-71) with the exception for $V_{\mathbf{p}_0 e}(t_3)$ where $\mathbf{p}_2$ was replaced with $\mathbf{p}_0$. Here $\mathbf{p}_0^{(2)}$ and $\mathbf{p}_f^{(2)} = \mathbf{p}_2$ are the initial and final drift momenta of electron 2's continuum transition. We decided to use these new symbols in order to stay consistent with the CQSFA model. The (2) hyperscript will be in the next few equations. The action is still separable into that of electron 1 and 2

$$S(\mathbf{k}, \mathbf{p}_1, \mathbf{p}_0, \mathbf{p}_f, t_1, t_2, t_3, t_f) = S_1(\mathbf{k}, \mathbf{p}_1, t_1, t_2) + S_2(\mathbf{p}_0, \mathbf{p}_f, t_3, t_f), \quad (97)$$

where the action for electron 1 is given by eqn. (73) and remains unchanged from the NSDI SFA model. The action for electron 2 is identical to that of the CQSFA (eqn. 41) with the obvious replacement of symbols $t_0 \rightarrow t_3$ and that now $\mathbf{p}_f = \mathbf{p}_2$. Following the tunnel-exit approximation it may be written as

$$S_2(\mathbf{p}_0, \mathbf{p}_f, t_3, t_f) = S^{tun}(\tilde{\mathbf{p}}, \mathbf{r}, t_3, t_{3r}) + S^{prop}(\tilde{\mathbf{p}}, \mathbf{r}, t_{3r}, t_f). \quad (98)$$

The next step is to formally apply the Saddle Point Approximations. This includes the Van-Vleck Gutzwiller method (eqn. 30) for the path integrals and the multivariable SPA (eqn. 27) for integral over the ionization times. The SPA transition amplitude of the Frankenstein therefore is

$$M(\mathbf{p}_1, \mathbf{p}_2) = -i \sum_s (2\pi)^3 \frac{V_{\mathbf{p}_0 e}(t_{3s}) V_{\mathbf{p}_1 e, \mathbf{k}g}(t_{2s}) V_{\mathbf{k}g}(t_{1s})}{\sqrt{(t_1 - t_2)^3 \det S''(\mathbf{k}_s, \mathbf{p}_1, \mathbf{p}_2, t_{1s}, t_{2s}, t_{3s})}} \sqrt{\frac{\partial \mathbf{p}_f(t_f)}{\partial \mathbf{r}_s(t_0)}} e^{iS(\mathbf{k}_s, \mathbf{p}_1, \mathbf{p}_2, t_{1s}, t_{2s}, t_{3s})} \quad (99)$$

$$\sim \sum_s e^{iS(\mathbf{k}_s, \mathbf{p}_1, \mathbf{p}_2, t_{1s}, t_{2s}, t_{3s})}.$$

Note that exact new forms of the determinant of the 6x6 Hessian $S''(\mathbf{k}_s, \mathbf{p}_1, \mathbf{p}_2, t_{1s}, t_{2s}, t_{3s})$ w.r.t t_1, t_2 and t_3 , and the stability factor $\sqrt{\frac{\partial \mathbf{p}_f(t_f)}{\partial \mathbf{r}_s(t_0)}}$ have not been evaluated. It is unclear if these remain identical to those of the original NSDI SFA and CQSFA ATI models. We decided to neglect prefactors the prefactors anyway in an effort to simplify the numerical integration. The S''_1 prefactor depends on t_{1s} and t_{2s} (calculated with the NSDI SFA for electron 1) and t_{3s} (calculated with the CQSFA ATI for electron 2).

Removing this prefactor ensures that transition amplitude is completely factorisable into the processes of electron 1 and 2. The goal was to investigate the broad fundamental changes the integration of the CQSFA would bring, and these changes should manifest themselves in the action and less so in the prefactors. The lack of the prefactors is indicated in the second line of the above equation.

Technically the above expressions should be evaluated in the limit $t_f \rightarrow \infty$. However for numerical approximation picking a reasonably high finite value e.g $20.25 \times \frac{2\pi}{\omega}$ a.u. (20.25 cycles) is sufficient. In the above equations the dependence of action S_1 on the final time t_f is ignored, which technically should not be the case due the presence of the time evolution operator $U_V^{(1)}(t_f, t_3)$. This should introduce a phase factor dependent on t_f that ensures that electron 1 retains the same relative phase to electron 2 throughout the continuum transition. This phase factor can be neglected if the difference between t_3 and t_f is an integer value of $\frac{2\pi}{\omega}$ because then electron 1's Volkov final state at t_f will have the same phase as at t_3 (consider $A(t_f) = A(t_{3s})$). Since electron 2 is likely to ionize around a peak ($0.25 \times \frac{2\pi}{\omega}$), any value of $(n + 0.25) \times \frac{2\pi}{\omega}$ should be sufficient for n integer. Furthermore we do not explicitly need to test for the causality condition $\Re t_2 < \Re t_3$ because we consider a continuous monochromatic electric field due to which any solution can simply be displaced by one cycle which will always satisfy causality.

The application of the Uniform SPA (eqn. 81) only affects the action of electron 1 but not that of electron 2:

$$M(\mathbf{p}_1, \mathbf{p}_2) = \left(e^{iS_1^{short}(\mathbf{p}_1)} + e^{iS_1^{long}(\mathbf{p}_1)} \right) \times \underbrace{\left(e^{iS_2^{one}(\mathbf{p}_2)} + e^{iS_2^{two}(\mathbf{p}_2)} \right)}_{M_2(\mathbf{p}_2)} \xrightarrow{USPA} M_1(\mathbf{p}_1) \times M_2(\mathbf{p}_2). \quad (100)$$

Here we also only consider the influence of orbit 1 and 2 for electron 2, as these remain similar to the SFA's original long and short orbits. Orbit 3 (and somewhat also 4) are associated with (soft) re-scattering

which physically represent a major departure from the RESI SFA model. The actions of electron 1 and 2, in the above notation are given by S_2^{one} and S_2^{two} respectively.

As indicated in the above equation, the transition amplitude is factorisable which allows the single cycle Partial Momentum distribution function to be expressed in terms of two independent integrals.

$$W(p_{1\parallel}, p_{2\parallel}) = \int |M_1(p_{1\parallel}, p_{1\perp})|^2 dp_{1\perp} \times \int |M_2(p_{2\parallel}, p_{2\perp})|^2 dp_{2\perp} = W_1(p_{1\parallel}) \times W_2(p_{2\parallel}). \quad (101)$$

which each represent a probability density function of electron 1 and 2. The two factors W_1 and W_2 represent probability density contribution to the PMDs of electron 1 and 2 respectively. Note that at this point we consider the two components in the x-y plane as just one component in the $\perp$ direction and use $dp_{\perp}$ instead of the formal $d^2p_{\perp}$ as in eqn. (2). Please note that when considering only one orbit for electron 2, the (integrand of the) probability density for electron two only depends on the imaginary part of the tunneling action $\Im S^{tun}(\mathbf{p}_2)$

$$|e^{iS_2(\mathbf{p}_2)}|^2 = e^{-2\Im S^{tun}(\mathbf{p}_2)} \times |e^{i[\Re S^{tun}(\mathbf{p}_2) + S^{prop}(\mathbf{p}_2)]}|^2 = e^{-2\Im S^{tun}(\mathbf{p}_2)} \quad (102)$$

where by construction the propagation action $S^{prop}(\mathbf{p}_2)$ is always real.

In summary we have considered several simplifications and assumptions through which the Partial Momentum distribution function (eqn. 100) can be factored into two terms $W_1(\mathbf{p}_1)$ and $W_2(\mathbf{p}_2)$ representing the probabilities of electron 1 and 2 respectively. This means we can re-use the existing NSDI and CQSFA C++ code to independently calculate the actions for electron 1 and 2 respectively, and then pipe their output together to calculate the PMDs.

Very clear summary

6.4 Interpolation and Coding

Unfortunately our initial goal of leaving the original codes unaltered quickly became unfeasible due to a few incompatibilities between the NSDI and CQSFA codes. Minor incompatibles like different physical definition of variables or different use of input field parameters were solved through simple fixes and unit converters. The most significant incompatibility were the different structures of the grid of finite momentum points at which the transition amplitudes are calculated. The CQSFA code uses a radial grid in which points are distributed along circles, i.e

$$(p_{\parallel}, p_{\perp}) = (R_n \cos(\theta_m), R_n \sin(\theta_m)) \quad R_n = n \frac{R_{max}}{N}, \quad \theta_m = m \frac{\pi}{M} \quad (103)$$

where $n = 0, \dots, N$ and $m = 1, \dots, M$ are integers and R_{max} is the maximum radius of momentum (typical value $3a.u.$). Note that this grid only considers the upper half plane (ie $0 \leq \theta \leq \pi$), as the both the transition amplitudes and action in ATI are symmetric about the $p_{\perp} = 0$ axis. The NSDI on the other hand uses a typical Cartesian grid along which points are spaced in rows and columns of constant spacing ΔP that fill a rectangle of dimensions $2p_{\parallel max} \times 2p_{\perp max}$, i.e

$$(p_{\parallel}, p_{\perp}) = (n\Delta P, m\Delta P) \quad \Delta P = \frac{2p_{\parallel max}}{N} = \frac{2p_{\perp max}}{M}. \quad (104)$$

code

To be able to effectively combine the two results we needed to ensure that both programs use the same grid to calculate the PMDs. We decided to redesign the NSDI code to use the radial grid of the CQSFA,

This is very nice and illustrative.

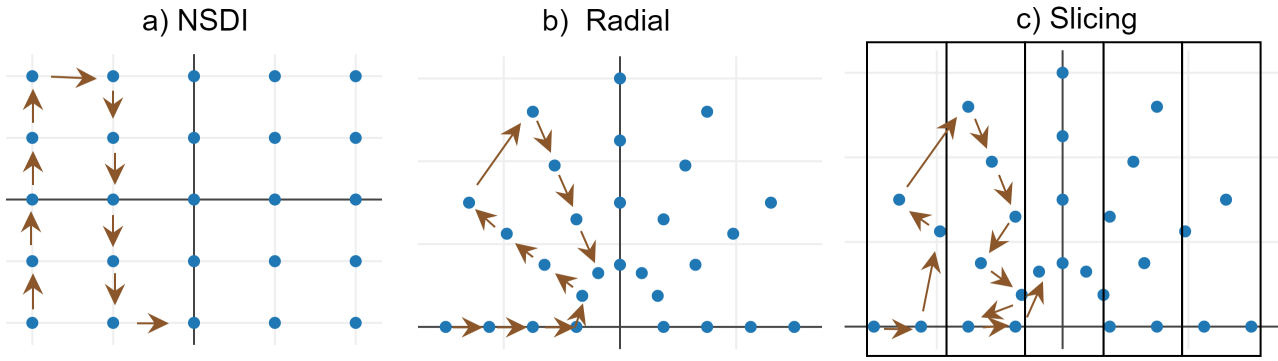


Figure 16: Representation of the convergence-path and grid structure in momentum space ($p_{\parallel}, p_{\perp}$ plane) for the NSDI, radial and the Slicing method. The brown arrows indicate the convergence-path used to evaluate the saddle point solutions, where the final solution at point is used as initial guess on the next one. Note that first guess usually is not the lower left point which here was used for visual convenience. In panel (c), the boxes indicate the slices into which each points are sorted. Convergence-path (b) is what we used in the end.

as redesigning the CQSFA code to adapt a Cartesian grid appeared difficult since it involves intricate procedures in which new momentum points are dynamically in the event of divergences. At the time we also had more experience with NSDI theory and code so we were much more confident in redesigning the NSDI code.

However applying a circular grid to ~~the~~ NSDI created two unique problems; the first getting the Saddle point solutions to converge and the second being able to interpolate the results.

A radial grid has a non-uniform density of points, as most points are concentrated at centre and thin out at the further away. For NSDI this is not ideal, as the classical region (in which the solution times (t_{1s} and t_{2s}) and action vary slowly) usually is located somewhat near the centre and the non-classical region (where the solution times and action vary very fast) is located further away. Ideally a higher concentration of points is desirable in the non-classical region to ensure that solution times for electron 1 converge. But this is precisely not the case for a radial grid. Initially the 'convergence-path' used by the CQSFA was employed for electron 1 (see fig. 16). Along the convergence path, the saddle point solution of one point is used as the initial guess for a neighbouring point. We initially employed a convergence path that is *similar* to that of the CQSFA²; the convergence-path moves radially until the centre or edge of the circle or the centre is reached in which it will switch to the next neighbouring radial line. See fig. 16b for a visualisation. However the saddle point times only converged for a very high number of points ($N \times M$) which ensure that the outer density in the non-classical regions is sufficiently high. Calculations with very high number of points are inefficient as most points are wasted near the centre where they are not needed for electron 1.

I thought maybe it was possible to increase efficiency by using a smarter 'convergence path' with which the saddle point times would converge at a smaller density of points. The idea was to emulate the convergence-path of the original NSDI in which the convergence-path moves vertically up or down the

²The CQSFA first solves the ionization times along a ring of $0.6\sqrt{Up}$ and then moves radially inward simultaneously

columns of constant $p_{\parallel n}$ until the edge is reached after which it switches to a neighbouring point. In this new method, all points were sorted into rectangular slices that vertically span from $-p_{\perp max}$ to $+p_{\perp max}$ and have a small horizontal width in the parallel direction as shown in fig. 16 and within each slice points are sorted with increasing or decreasing perpendicular momentum. This largely imitates the original path of the NSDI solver, and successfully lead to convergence at lower number of total points. I speculated that reason for why this path would get away with this despite still having a lower density of points at the edge, is that at constant $p_{\parallel}$ solution times and action vary slower w.r.t changing $p_{\perp}$ than path with constant $p_{\perp}$ and changing $p_{\parallel}$.

I also figured that this slicing method could also be used to tackle the other problem, which was the interpolation needed to calculate the probability density function $W_{\mu}(p_{\mu\parallel})$ of each electron ($\mu = 1, 2$). To calculate $W_{\mu}(p_{\mu\parallel})$, one must take the line of constant $p_{\mu\parallel}$ and integrate the square of the magnitude of the transition amplitude along $p_{\mu\perp}$. For the Cartesian grid of the original NSDI this is pretty easy as points in a column conveniently have constant $p_{\mu\parallel}$, meaning even simple 1D numerical interpolation schemes like Trapezium rule can be used. In the new radial grid this is no longer the case meaning that 2D-interpolation-integration schemes have to be used to prepare sets of points of constant parallel momentum, which can be integrated over.

The initial hope was to use this slicing method to approximate this interpolation, by considering $W_{\mu}(p_{\mu\parallel})$ as a weighted sum of all actions within each slice. However further experimentation showed that this yields 'fuzzy' transition amplitudes. Only by considering a high number of total points could the slices be thin enough and contain a sufficient number of points to paint a reliable PMD picture. Since the goal of this slicing method was to reduce the total number of points, this idea was quickly disbanded in favor of developing a sophisticated interpolation-integration method.

The first hope was to use polynomial-interpolation. Polynomial-interpolation methods interpolate an original a function $f(x)$ whose value is only known at a set of $N + 1$ discrete points x_n , ie $f_n = f(x_n)$. Polynomial interpolation methods create a weighted N degree polynomial $p(x) = a_0 + a_1x^1 + a_2x^2 + \dots + a_Nx^N$ that approximates the original function. This weighting takes all values of $f(x_n)$ across the entire grid into account, but locally still retains the original values at the discrete points ie $p(x_n) = f(x_n)$. A 1D example of polynomial interpolation is Legendre interpolation

$$p(x) = \sum_{n=0}^N l_n(x) f_n \quad l_n(x) = \prod_{m=0, m \neq n}^N \frac{x - x_m}{x_n - x_m} \quad (105)$$

as explained here [57]. Because the interpolation output is a polynomial, the integration not only becomes trivial ($P(x) = \int p(x)dx = a_0x + \frac{a_1}{2}x^2 + \dots + \frac{a_N}{N+1}x^{N+1}$) but also can be evaluated at any new arbitrarily chosen value of x' even after the interpolation procedure is complete. This is not possible with most other interpolation methods, which usually interpolate the function at a specific x' , ie choosing another x' means the interpolation procedure has to rerun again. Furthermore it is possible to construct a vector $\vec{l}(x)$ whose elements only depends on the distribution of discrete points x_n (independent from $f(x_i)$) whose dot product with the vector $\vec{f}$ of elements $f(x_i)$ gives the polynomial. ie $p(x') = \vec{l}(x') \cdot \vec{f}$. This effectively presents a 'golden hammer' approach; we just need to calculate a set of vectors $\vec{l}(x')$ once for the radial grid, which then can be easily applied to all points at once in order to interpolate actions.

1402
fuzzy

This efficiency becomes beneficial when considering multiple channels and cycle events, for which each combination each four interpolations are required (2 orbits for each of the 2 electrons). Using standard interpolation methods need to be completely re-applied over and over again. *Meaning: you had to change things for e_2 or well.* Polynomial-integration methods are very well understood in 1D in the form of Lagrange polynomials and easily extended to 2D systems of rectangular meshes (ie set of points $(x_n, y_m) = (h(n), g(m))$ where h and g are functions that act on integers n and m respectively). This unfortunately is not the case for radial grid. My initial efforts was to introduce a radial Lagrange polynomial approach introduced here [58], however this only works with radially symmetric functions. The next approaches focused on using a modified version of Birkhoff interpolation introduced in [59]. For more information and in depth discussion on 2D polynomial interpolation see [60]. However this approach become more complicated, and the integration no longer was trivial, nullifying most the advantages outlined above. It also became clear that lower order interpolation schemes like nearest-neighbour schemes became more appropriate due to very fast moving nature of some the integration functions.

We therefore abandoned this 'golden hammer' polynomial interpolation approach and instead focused on a simple yet proven method; existing open-source polynomial interpolation tools of SciPy (Python) package. In term's of work contribution; I have prototyped the slicing method for solution times and integration in Mathematica, as conducted the literature survey into polynomial interpolation. The idea was to test these features first in Mathematica (which has much quicker development speed and better visualisation but lower operating efficiency than C++) and then for Gergely Eöry to incorporate these into the new C++ NSDI code. The new C++ NSDI code and python integration-interpolation was solely coded by him and more details can be found in his MSci thesis.

This solution however is not perfect as a very high number of momentum points is needed to ensure that the density on outer regions of the radial grid is sufficiently big. Because of this the slicing method for the convergence-path became redundant and was replaced with original radial convergence-path as shown in fig. 16b. Another issue that was we initially did not fully anticipate was the fact the a only square region within the circular grid can actually be used for integration. This has to be done to ensure that the contribution at each parallel momenta $p_{\parallel}$ is fair; otherwise for $p_{\parallel}$ near the centre of the circle there is higher range of $p_{\perp}$ to integrate over compared to $p_{\parallel}$ values at edge of the circle. This means that overall size of the radius of the momentum grid has to be unnaturally large. This among other issues explained in Gergely Eöry's thesis, meant that the NSDI and SFA both produced use-able output only in as mall specific range of field and ionization parameters (ω, I, I_p) . Also getting the interpolation to work was difficult. The set of parameters closest to the NSDI regime was $(\omega, I, I_p) = (800 \text{ nm}, 4 \times 10^{13}, 0.6115 \text{ a.u.})$. The value $I_p = 0.6115 \text{ a.u.}$ corresponds to the $4P$ excited state configuration of Argon³. For electron 2 this constitutes Keldysh parameter of 1.66889 which is in the low end of the multiphoton ionization regime [61]. All results discussed in this thesis are in the context of this set of parameters.

³See <https://physics.nist.gov/PhysRefData/Handbook/Tables/argontable6.htm> and conversion 1 a.u. = $2.1947463 \times 10^5 \text{ cm}^{-1}$ for other possible values.

High and, I would say. The parameters are tricky and we need to perform further tests to

ensure whether RESI is the dominant mechanism.

7 Results, Analysis and Discussion

This section is very nice and highlights the effort heavily

Presented in this section are all results and analysis. Section 7.1 contains the Partial Momentum Distributions of the Frankenstein that were calculated using the interpolation method. Section 7.2 contains plots analysis of the initial probability distributions of electron 2 upon ionization. In Section 7.3 the relation between the initial and final momentum is investigated in an effort to explain the origin of new features identified in section 7.1. Methods and analysis presented in the last two subsections are my own work.

to make it work

7.1 Partial Momentum Distributions

The output of the interpolated Partial Momentum Distributions of the Frankenstein for the aforementioned set of parameters is shown in figure 17, which also includes the PMD of the old NSDI SFA model. A few changes and new features can be identified. The overall change is that most transition probabilities appear to be located closer towards the axes. There is also less intersection between the two main regions. A specific new feature that can be found in the single orbit PMDs for orbit 1 and 2 of electron 2, is a small momentum suppression near the axes. In orbit 1 (fig. 17b) this suppression continues along axes, creating a small ridge. This ridge becomes more apparent in fig. 18, which shows W_2 , the probability density contribution term to the PMD (see eqn. 101) of electron 2. Please note that there is a asymmetry in fig., whose origin is unknown. We believe that is an artifact that stems from the interpolation, as the original transition amplitudes are symmetric. In orbit 2 (fig. 17c) the suppression can only be found near the axes as the probability peaks directly on the axes, forming two smaller and more subtle ridges. Both these features become more evident by looking at the squared Transition Amplitude (figures 17e-f) for orbit 1 and 2 respectively. It should be noted that the peak along the axes in Orbit 2 may be an artifact that arises due to the lack of prefactors. For instance, a comparison of fig. 1 to fig. 3 in [47], where squared transition amplitudes are plotted for similar field parameters, indicates that the introductions of prefactors dampens this peak near the axis. The combination of orbit 1 and 2 for electron 2 leads to 'ripples' parallel to the axes. Near the origin these intersect, leading to a weak interference signal. [These 'ripples' are likely to be a remnant of the "fan-shape structure" found in (H)ATI, as can be seen in fig. 7 of [47].] This is interesting because it would be another

we need to think about how to test this type further.

of interference that does survive.

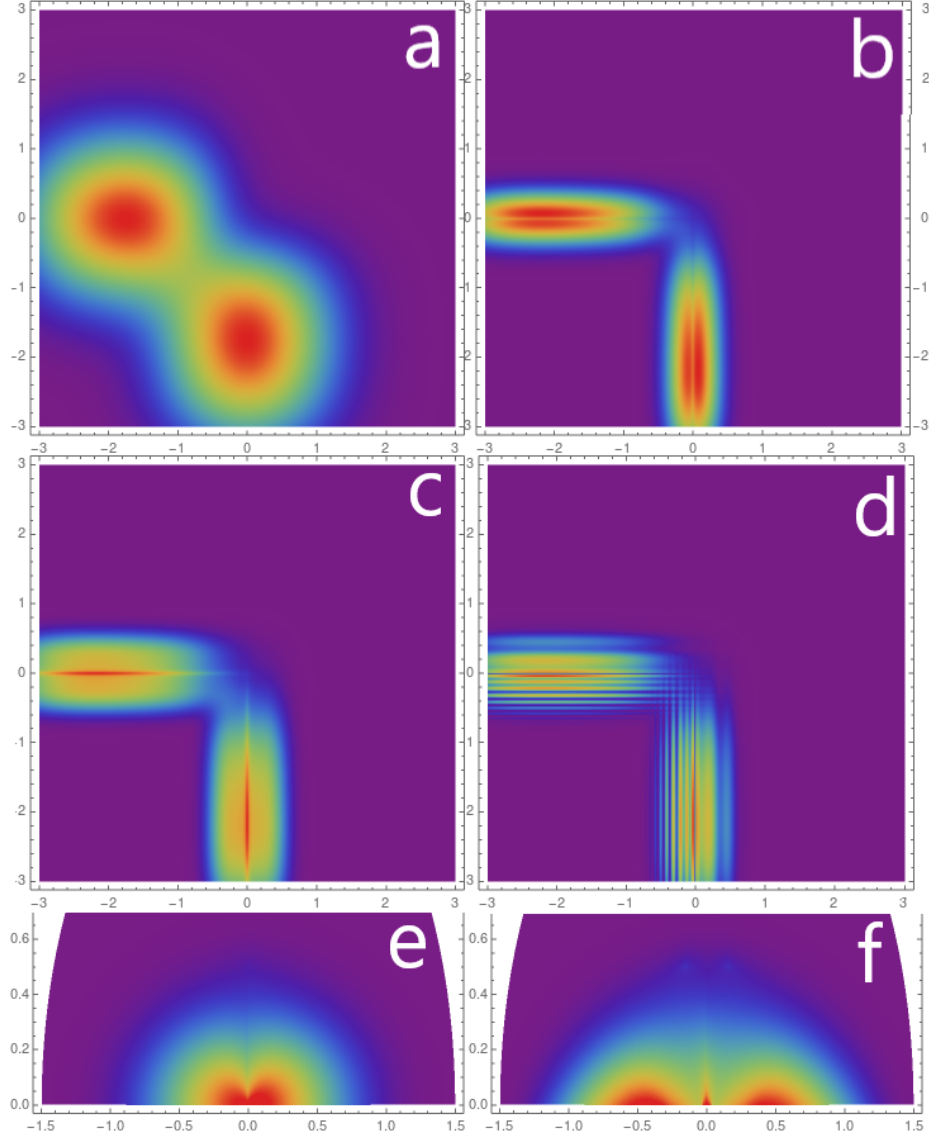


Figure 17: Single cycle single channel Partial Momentum Distributions (panels (a) to (d)) and squared Transition Amplitudes (panels (e) and (f)) using the Interpolation method. Panel (a) is combined orbit 1 and 2 PMD according to the old NSDI SFA. Panels (b) and (c) are single orbit for orbits 1 and 2 using the Frankenstein respectively. Panel (d) shows the PMD for combined orbit 1 and 2 according to the Frankenstein. All panels use the same combined long and short orbit contribution according to the USPA for electron 1. For all PMDs, the exchange in momenta $\mathbf{p}_1 \leftrightarrow \mathbf{p}_2$ was taken into account. Panels (e) and (f) show the CQSFA squared transition amplitudes for orbit 1 and 2 of electron. Horizontal and vertical axis are the (final) parallel momentum for electron 1 $p_{1\parallel}$ and electron 2 $p_{2\parallel}$ respectively, all in units of $\sqrt{Up}$. Parameters: $(\omega, I, I_p) = (800 \text{ nm}, 4 \times 10^{13}, 0.6115 \text{ a.u.})$. Courtesy of Gergely Eöry.

7.2 Initial Probability Distributions

My next goal was to derive new momentum constraints for the Frankenstein that could broadly predict these features. Since electron 1 is calculated by the SFA, the contribution of electron 1 to the PMDs as predicted by eqn. (91) remains valid. All new features are therefore caused by electron 2 and are encoded in W_2 . This turned out to be significantly surprisingly difficult due to the shift in initial parallel momentum $p_{0\parallel}$ to final parallel $p_{f\parallel}$ momentum that occurs in the CQSFA, and shall be referred to as

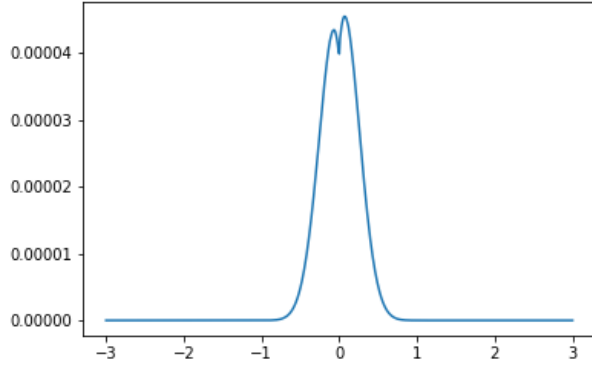


Figure 18: The normalized contribution of electron 2 to the PMD (the $W_2(p_{f||})$ term in eqn. (101)) for orbit 1 in terms of the final momentum $p_{2||}$ of electron 2 in units of $\sqrt{U}p$. Parameters: $(\omega, I, I_p) = (800 \text{ nm}, 4 \times 10^{13}, 0.6115 \text{ a.u.})$. Courtesy of Gergely Eöry.

the 'momentum-mapping'. Predicting the final value $\mathbf{p}_f$ based on the knowledge of the initial value $\mathbf{p}_0$ is difficult as it requires solving Newton's classical equations of motion (eqns. 44, 45 and 48), which have no known analytic solution and constitute to a non-conservative system in which energy is not conserved. Predicting even the maximum possible value of $p_{f||}$ for a given $\mathbf{p}_0$ is problematic, as it is difficult to know whether a given value $\mathbf{p}_0$ belongs to nonphysical diverging orbit. Solving the problem backwards (ie predicting $\mathbf{p}_0$ for a given $\mathbf{p}_f$) analytically is even more difficult since it involves solving the SFA with perturbatively increasing C .

I therefore wanted to investigate to what extent the momentum-mapping plays a role in forming the shape of the final PMD patterns and whether it can be neglected. To achieve this, I wanted to compare the initial and final partial probability distributions of electrons 2. Note that for single cycle single orbit distributions only the imaginary part of the Tunneling action S^{tun} (eqn. 49) contributes to the overall probability, meaning that once the electron has entered the continuum, the transition amplitude of an orbit does not change. This means that any shifts in the probability distributions are purely caused by shifts in the momenta. Using the numerical output of the CQSFA to plot the probability distributions in terms of initial momentum is not straightforward, as the CQSFA is a backwards method and therefore has unstructured grids of initial momenta, also requiring interpolation. Luckily the tunneling action S^{tun} can easily be expressed purely as a function of $\mathbf{p}_0$, which led me to creating a simple Mathematica script that calculates the initial transition amplitude and probability distributions. For this I used a structured Cartesian grid of initial momentum points (similar to that of the original NSDI SFA) with which numerical integration needed to calculate W_2 became trivial. This was done for orbit 1 and 2.

The analytic expressions developed for the CQSFA in [47] were extremely helpful. The tunneling action can be divided into an SFA-like term and a Potential Barrier term I_V . The SFA like term is known fully analytically in terms of the (complex) ionization time t_0 which in turn can be expressed in terms of $\mathbf{p}_0$ (see eqns. 53 and 54). The potential barrier term I_V can be calculated numerically or approximated analytically with a regularized expression that removes its divergent components (see eqn. 46 in [47]). To be consistent with the literature [47], I have used the analytic approximation for all plots shown in

this thesis, but it should be noted that this approximation overestimates the height of the actual peaks.

The results of the initial probability distributions for orbit 1 and 2⁴ are shown in figures 19 and 20 respectively.

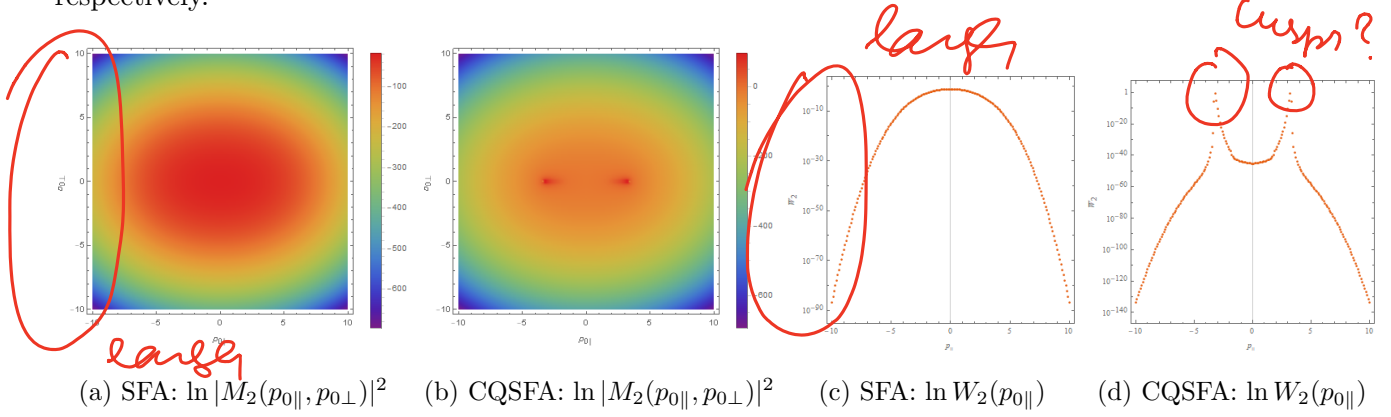


Figure 19: Orbit 1 *initial* squared Transition Amplitude and Partial Momentum Contributions of electron 2 for the CQSFA and SFA. Panels (a) and (b) show the logarithmic un-normalized transition amplitudes in terms of initial parallel $p_{0\parallel}$ and perpendicular $p_{0\perp}$ momentum. Panels (c) and (d) show the normalized partial Momentum Distribution W_2 of electron 2 in terms of initial parallel $p_{0\parallel}$ momentum. All momenta in units of $\sqrt{U_p}$. Parameters: $(\omega, I, I_p) = (800 \text{ nm}, 4 \times 10^{13}, 0.6115 \text{ a.u.})$.

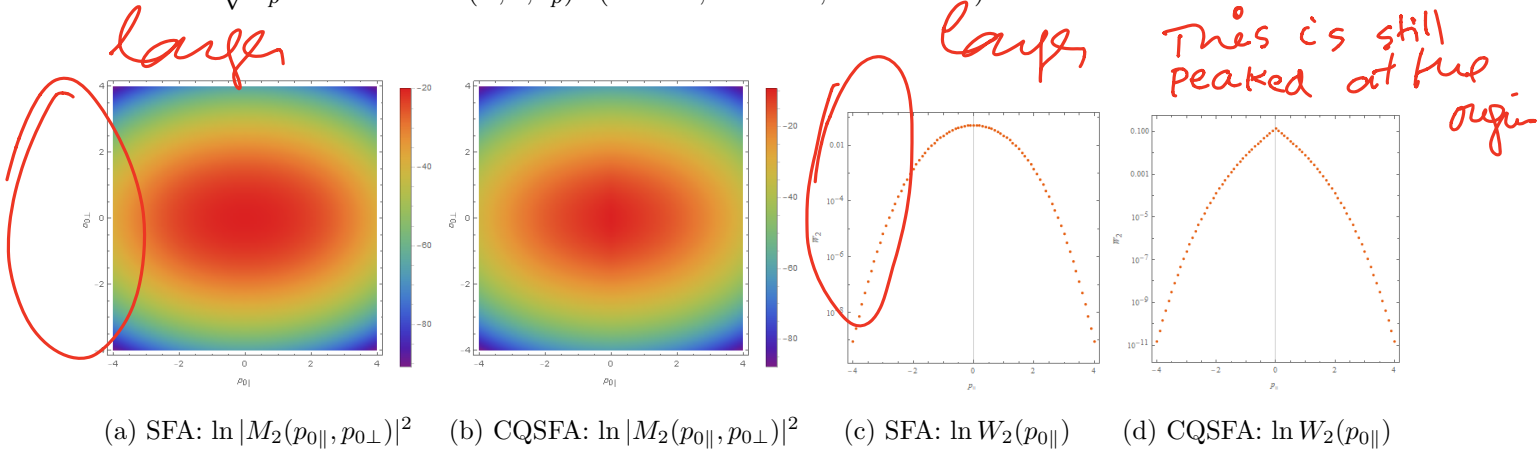


Figure 20: Orbit 2 *initial* transition amplitudes and Partial Momentum Contributions of electron 2 for the CQSFA and SFA. Panels (a) and (b) show the logarithmic un-normalized transition amplitudes in terms of initial parallel $p_{0\parallel}$ and perpendicular $p_{0\perp}$ momentum. Panels (c) and (d) show the normalized partial Momentum Distribution W_2 of electron 2 in terms of initial parallel $p_{0\parallel}$ momentum. All momenta in units of $\sqrt{U_p}$. All momenta in units of $\sqrt{U_p}$. Parameters: $(\omega, I, I_p) = (800 \text{ nm}, 4 \times 10^{13}, 0.6115 \text{ a.u.})$.

For Orbit 1, a significant difference between the standard SFA and the CQSFA can be seen. As seen in fig. 19b, two 'horns' emerge in the square of the transition amplitude (fig. 19b) near $(p_{0\parallel}, p_{0\perp}) \approx (\pm 2.5 \times \sqrt{U_p}, 0)$. These peaks are manifestations of the potential barrier term I_V whose omission leads the probability distributions looking similar to the original SFA term. The two 'horns' explain why transition probability peaks next to the core lead to a slight momentum suppression at $p_{0\parallel} = 0$. This suppression is clearly found in the probability density contribution W_2 of electron 2 (fig. 19d), and looks

⁴I have used eqn. 54 to generate ionization times for orbit 2, but this eqn. is also applicable to orbits 3 and 4. As the shift in position vector and shift in momenta was not taken into account, this technically means that orbits 2, 3 and 4 together are combined here.

somewhat similar to that of the final distribution obtained from the interpolated CQSFA (fig. 18). The main difference between the initial and final shapes of W_2 is that the former being more pronounced and spreading wider than the latter. I am not certain whether this over-pronouncement of the ridge may come from using the analytic approximation for I_V .

In orbit 2, the probability distributions of the CQSFA look quite similar to the original SFA. The most notable difference is that the CQSFA has increased transition probabilities near p_{0par} . This is clearly seen in probability density contribution W_2 (fig. 20d) of electron 2, where the CQSFA is significantly 'sharper' near the peaks. This also matches the final probability distributions of the interpolated CQSFA (fig. 17), which peak at the axes. However, notably absent is the momentum suppression near $p_{0\parallel}$ that forms the two ridges in the final distributions. This feature must therefore arise through the momentum-mapping.

Very likely. We need to have a closer look over the exams are over.

7.3 Momentum-Mapping

To establish further insight, my next goal was to employ a 'shooting' method to numerically investigate how initial momenta translate to their final ($p_{0\parallel}, p_{0\perp}$) parallel momentum $p_{f\parallel}$ in an effort to identify which specific regions of the initial probability distributions experience a shift. To do this, I have employed Mathematics's built-in numerical solving tool (for partial differential equations to solve Newton's classical equations of motion) for a set of cycles comparable to that of the CQSFA. These results for same parameters as the interpolated CQSFA are shown in figures 21 and 22 for orbit 1 and 2. Note that the orbits were not collected into their respective orbit types by formally checking if the initial and final position and momentum vectors match the orbit definitions as outlined in section 4.4. I simply used equations (53) and (54) to generate the ionization times (and thus all other initial boundary conditions that govern an orbit's trajectory) for orbits 1 and 2 respectively. This technically means that other possible orbits are embedded in these results. This is not too much of a concern, since the forward shooting method might anyway not the exact same problem as the CQSFA's backwards method. The aim was simply to investigate how regions of the initial and final probability distributions relate to each other.

This is more extreme for the other 2 orbits - see 2nd

For orbit 1 in regions away from the origin, the initial and final parallel momenta are remarkably similar (see fig. 21a) which matches the predictions of the SFA model. Closer to the centre two somewhat circular regions form on the parallel axis between $0 < p_{0\parallel} < 3\sqrt{U_p}$ and $-3\sqrt{U_p} < p_{0\parallel} < 0$ (ie roughly located between the 'horns' of barrier potential term I_V) with maximum widths of roughly $-\sqrt{U_p} < p_{0\perp} < \sqrt{U_p}$. In these regions final momenta strongly deviates from the initial momenta and become 'chaotic'. This includes many points constituting unphysical orbits that diverge (see the white spots in fig.21b) or orbits whose final parallel momentum is sufficiently outside the 'box' of the final momentum grid considered in the CQSFA. This means the transition amplitudes of orbits whose initial momenta lies in these 'chaos' regions are distributed throughout the final momentum space and are therefore less pronounced in the final momentum space. This could potentially explain why the final momentum ridge suppression in fig. 18 is less pronounced than the initial ridge in 20 that is formed by barrier term I_V . The initial ridge is completely embedded in these 'chaos' regions, and thus 'washed' out. To better highlight which initial regions stay unaffected by the momentum-mapping, I have plotted the relative change in parallel

2018 J. Phys. B.

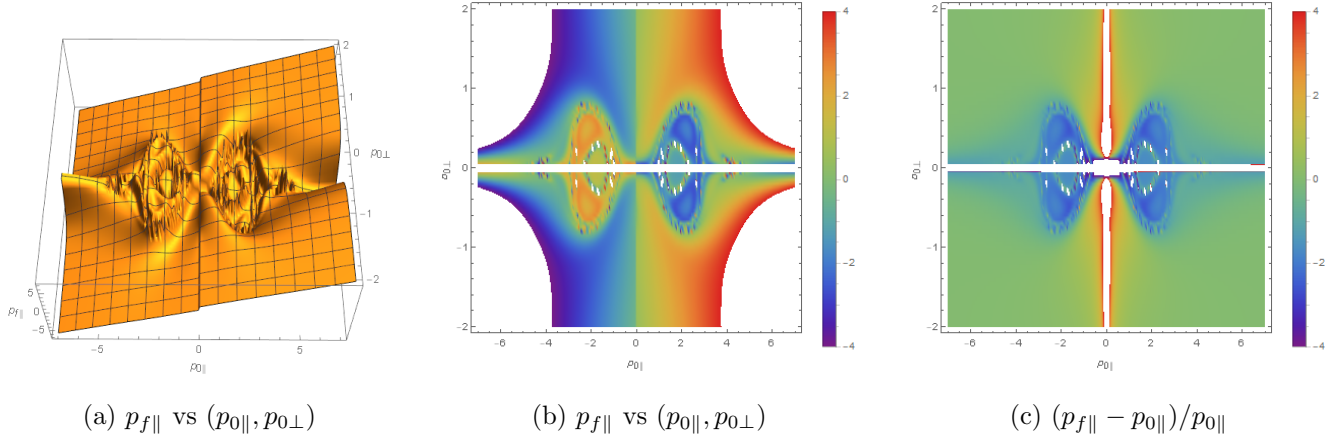


Figure 21: Orbit 1 Momentum-mapping. Panels (a) and (b) show the final parallel momentum $p_{f||}$ in (in terms of initial parallel $p_{0||}$ and $p_{0\perp}$ (x and y axis) momentum as a 3D and density plot respectively. Only range $-4\sqrt{U_p} < p_{f||} < 4\sqrt{U_p}$ is plotted. Panel (c) shows the relative change in momentum w.r.t to the original. Green regions indicate $p_{f||} \approx p_{0||}$ in agreement with the SFA, other colors indicate a strong change in the momentum. Any white regions in (b) and (c) indicate orbits that will diverge or whose final momentum is outside the 'box'. All momenta in units of $\sqrt{U_p}$. Parameters $(\omega, I, I_p) = (800 \text{ nm}, 4 \times 10^{13}, 0.6115 \text{ a.u.})$.

momentum $(p_{f||} - p_{0||})/p_{0||}$ as density plot (fig. 21c). Any points where $p_{f||} \approx p_{0||}$ are indicated in green (zero relative change) and behave similar to the SFA. Other colours indicate a strong change in momentum that deviates from the SFA. Note that the parallel momentum axis contains unphysical divergent orbits, as here the initial perpendicular momentum remains small enough that electron is driven into the centre of the core as explained in the next subsection. The divergence along the vertical perpendicular axis shown in fig. 21c is likely an artifact that arises due to non-continuity at $p_{0||} = 0$ of the orbit definition. For orbit 2 a similar 'chaos' region can be found near the origin (see fig. 22a and b). Compared to orbit 1 these regions are small, located much closer to the origin. They appear as circular rings. The final parallel momentum in these regions appears more chaotic and contains more non-physically diverging orbits. Interestingly, between the ringlike 'chaos' structures, a small non-chaotic island is formed around the origin that extends along the perpendicular axis. This explains the existence of the two small ridges and momentum suppression that can be found parallel and close to the axes in the final probability distributions (see fig. 17) but not in the initial distributions. Unlike orbit 1, the region containing non-physical diverging orbits along the parallel axis (see fig. 22b) is contained within the island of chaos. Similarly to orbit 1, the divergence along the perpendicular axis in fig. 22c is an artifact that arises from the non-continuity in the orbit definition.

My initial hypothesis was that in these 'chaos' regions, the initial strength of the Coulomb potential $V_c = -C/z_0$ (where z_0 is the tunnel exit in eqn. 48) is particularly strong because for a vanishing strength $C \rightarrow 0$ the CQSFA collapses to the SFA. However, a plot of V_c w.r.t initial momentum (see fig. 23) shows whilst the strength of the Coulomb potential peaks near the origin it however does not share any resemblance to the specific structures of the 'chaos' regions. There must therefore be other factors that play a role in the formation of these regions. I therefore ~~sought~~ ^{sought} further insight into the ~~trajectories~~

large labels

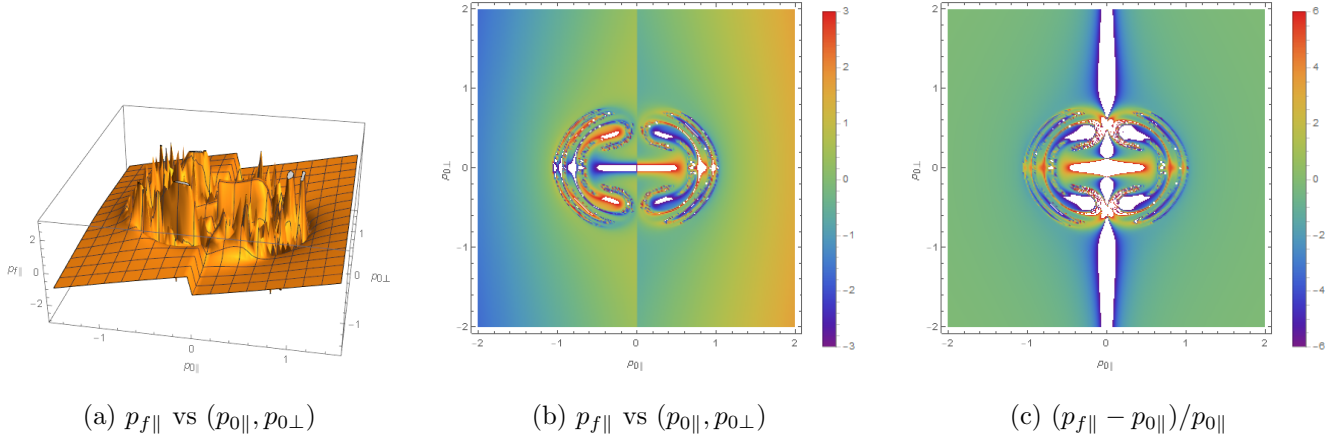


Figure 22: Orbit 2 Momentum-mapping. Panels (a) and (b) show the final parallel momentum $p_{f||}$ in (in terms of initial parallel $p_{0||}$ and $p_{0\perp}$ (x and y axis) momentum as a 3D and density plot respectively. Only range $-13\sqrt{U_p} < p_{f||} < 3\sqrt{U_p}$ is plotted. Panel (c) shows the relative change in momentum w.r.t to the original. Green regions indicate $p_{f||} \approx p_{0||}$ in agreement with the SFA, other colors indicate a strong change in the momentum. Any white regions in (b) and (c) indicate orbits that will diverge or whose final momentum is outside the 'box'. All momenta in units of $\sqrt{U_p}$. Parameters $(\omega, I, I_p)=(800 \text{ nm}, 4 \times 10^{13}, 0.6115 \text{ a.u.})$.

of orbits with the goal to potentially identify parameters that can predict whether an orbit remains SFA-like, experiences a shift in momentum or non-physically diverges. This investigation was done by plotting a variety of different parameters including the position vector $\mathbf{r} = (r_{||}, r_{\perp})$, momentum $\mathbf{p}$, the kinetic energy T , the strength of the Coulomb potential V_c , the strength of the field's vector potential V_c w.r.t. to time t throughout an electron's journey in the continuum. Some of the prominent observations are outlined below.

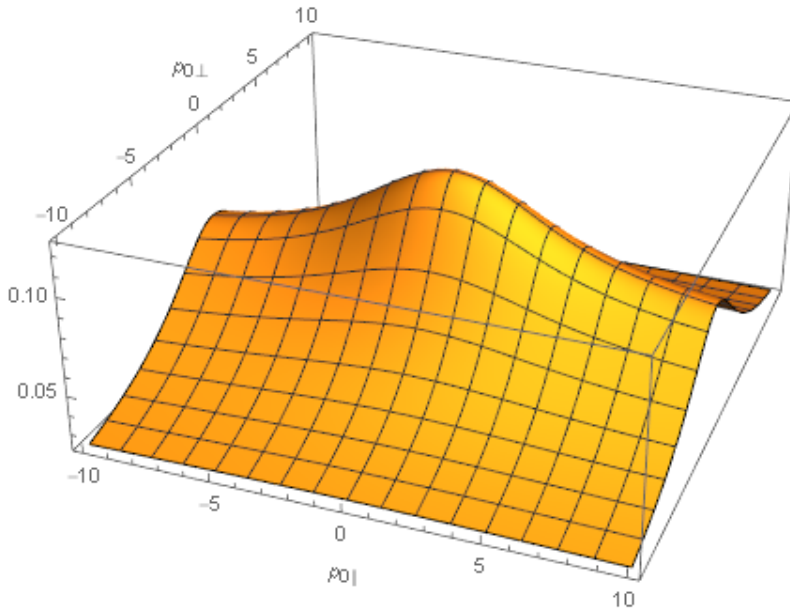


Figure 23: The initial strength of the Coulomb potential term $V_c = -1/z_0$ (a.u.) for Orbit 1 in terms of initial parallel and perpendicular momentum $p_{0||}$ and $p_{0\perp}$ in units of $\sqrt{U_p}$. Plotting V_c for orbit 1 yields an identical graph. Parameters: $(\omega, I, I_p)=(800 \text{ nm}, 4 \times 10^{13}, 0.6115 \text{ a.u.})$.

large labels

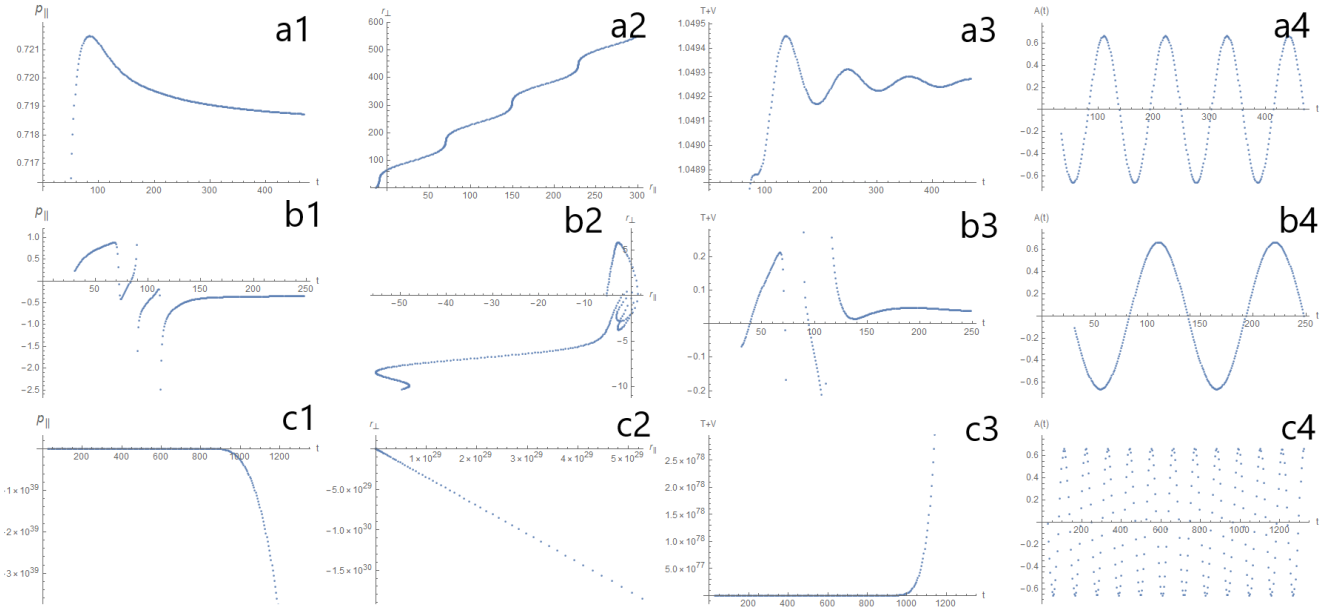


Figure 24: Trajectory data of 3 selected orbits. The initial momentum $(p_{0\parallel}, p_{0\perp})$ for the orbits rows (a), (b) and (c) are $(2.0, 4.0)$, $(0.7, 1.3)$ and $(0.1, 0.1)$ respectively and all in units $\sqrt{U_p}$. Panels in column 1 show the parallel momentum $p_{\parallel}$ as a function of time t . Panels in column 2 show the position along the perpendicular axis $r_{\perp}$ w.r.t position along the parallel axis $r_{\parallel}$. Panels in column 3 show the sum of the kinetic energy T and the Coulomb potential V in terms of time t . Panels in column 4 show the amplitudes of the field's electric vector potential $A(t)$ in arbitrary units w.r.t time t . Unless specified, all units are a.u.. Parameters: $(\omega, I, I_p) = (800 \text{ nm}, 4 \times 10^{13}, 0.6115 \text{ a.u.})$.

In orbits that are SFA-like (see fig. 24a for a example), the electron's parallel momentum initially experiences a small shift after which it oscillates in sync with the vector potential $A(t)$ during which the drift momentum remains almost constant. Interestingly, the majority of the shift occurs rapidly within a time-frame shorter than the period of $A(t)$. Similarly, the sum of the kinetic energy and coulomb potential $T + V$ only shifts rapidly in the beginning and after the first cycle it appears to oscillate with a diminishing amplitude. This also appears to be conserved at the crossings of $A(t)$. This reaffirms the SFA's main assumption that the electron initially only interacts with the binding potential (neglecting the external field) and afterwards only oscillates as a plane-wave Volkov state (neglecting the binding potential) but through continuum trajectory perspective. For a sufficiently large initial momentum, the interactions with binding potentially happen too rapidly before the time-dependent external field can meaningfully affect the momentum.

Orbits that experience a shift in parallel momentum (see fig. 24b for example), usually have a small tunnel exit z_0 and therefore have a strong initial value of V_c . If the initial momentum is sufficiently small, the Coulomb potential will trap the electron inside the barrier and force it to cycle around the nucleus, similar to satellites orbiting a planetary or stellar object. However the intervention of the vector potential of the external field continually shifts the positions of these trajectories. After some time, the configuration of the position and momentum vector and the phase ωt of $A(t)$ just become right for the field to push the electrons very close to the core. If the distance closest approach to core (the "periapsis" in astronomical terminology) becomes sufficiently low, the electron gains a significant boost in potential

energy and breaks out of potential barrier with a very high momentum. Afterwards the electron will oscillate like a plane wave. This is reminiscent of the 'slingshot maneuver' used for satellites to gain extra velocity around a orbiting object. The role of this effect in NSDI has been investigated in [56]. However this source claims that this effect leads to a new NSDI mechanism (named "slingshot ionization") that is independent to RESI. However this effect might actually be embedded in RESI (and other SFP phenomena like (H)ATI). A further investigation of this effect could be interesting. *I expect it to be present in orbit 2 already.*

Note that it is difficult to predict the magnitude and direction of momentum with which the electron escapes the ion core, as it is difficult to determine the time when all conditions for an electron to be driven to the core are satisfied. The smaller the distance of closest approach, the larger the escape velocity. However, I have identified that the initial perpendicular momentum needs to be sufficiently high to drive the electron into a circular orbit. If it is too low, then the electron is driven directly towards the origin, in which case the potential energy of the Coulomb force diverges, leading unphysical momentum. That is why trajectories with initial momentum near the parallel axis in fig 21b diverge. The trajectory of such an orbit is shown in fig. 24c. *This would be some kind of backscattered trajectory, I guess. We need to have a closer look at all this later.*

8 Conclusion and Improvement

This project has achieved the goal of calculating the PMDs for the new Frankenstein model. However, the outcomes that could be predicted are for a small set of field and ionization parameters, and only for single-cycle interference, which is more limited than initially aimed. A major limitation is that the calculation of the event interference critical in shaping the NSDI interference patterns could not be done. Additionally, further work is needed to improve the efficiency of the code which is currently limited by the requirement of high resolution momentum points.

Most of these issues follow from the difficulty to perform the interpolation and integration across the radial grid. I believe that if we would have initially decided to adapt the CQSFA code to work with a Cartesian grid (rather than adapting the NSDI SFA code to work with a radial grid) we would have had fewer issues getting the interpolation to work and therefore more time to also get event interference working. At the time, however, we chose not to modify the CQSFA code. I, therefore, recommend that any further work should focus on adapting the CQSFA to work with a Cartesian grid, as this would require a significantly lower density of momentum points. Further work also needs to be done to ensure that the NSDI and CQSFA code work with similar field and ionization parameters.

The main new features identified with the achieved results were a momentum suppression at the axes for orbit 1 and a smaller momentum suppression next to the axes for orbit 2. Through calculating the initial probability distributions, I identified that the momentum suppression in orbit 1 exists in the initial momentum distribution but is likely washed out through the momentum-mapping. The momentum suppression of orbit 2, however, only arises due to the momentum-mapping. This provides an opportunity to derive new momentum constraints for orbit 1 based on the analytic expressions for the initial tunnelling action which avoids the issue of the momentum mapping. However, these conditions would only work for single-cycle orbits because the effect of the propagation action would have to be taken into account for multiple cycles, which makes these conditions inapplicable on a multi-cycle interference. This could not

be investigated further due to time constraints.

The prefactors terms in eqn. (99) should also be incorporated as these might limit the momentum peak at across the axis in orbit 2 as described in section 7.2.

Another direction to expand upon would be developing better conditions to predict the 'chaos' regions in the initial momentum space in which orbits diverge or experience a significant change in momentum. The extent to which these regions are incorporated in the backwards momentum mapping of the CQSFA should also be formally investigated. The existence of effective conditions predicting the regions in momentum space where either the coulomb potential or the laser field dominates could allow developing a modified method in which perturbations w.r.t to the binding potential or the external field are considered in different regions. A similar concept has already been explored in [62]. Investigating whether 'sling-shot' acceleration is embedded in RESI is another possible direction to go on.

we need to have a closer look at what is going on.

Clear idea of the issues to be faced and how
Acknowledgement to push the problem further.

I would like to thank Gergely Eöry for being a good project partner. His contribution was coding the new C++ NSDI SFA code and python interpolation-integration code. I would also like to thank my supervisor Prof. Dr. Carla Figueira De-Morrison Faria for her close support. It has been a pleasure working with both of them. Dr. Andrew S. Maxwell, the author of the CQOrbit code, has been helpful his support and advise. His Doctoral thesis [18] has been instrumental in understanding both the theory of the CQSFA and RESI SFA. Likewise the MSci thesis [17] of Ahmed Al-Jawahiry and the Doctoral thesis of Tahir Shaaran have been helpful in understanding the SFA theory of (H)ATI and the SFA. Finally, I would like to thank Jakub Zalesak and Ulrich Holtmann for English language corrections in this thesis.

References

- [1] A. l’Huillier et al. “Multiply charged ions induced by multiphoton absorption in rare gases at 0.53 μm ”. In: *Phys. Rev. A* 27 (5 May 1983), pp. 2503–2512. DOI: 10.1103/PhysRevA.27.2503.
- [2] B. Walker et al. “Precision Measurement of Strong Field Double Ionization of Helium”. In: *Phys. Rev. Lett.* 73 (9 Aug. 1994), pp. 1227–1230. DOI: 10.1103/PhysRevLett.73.1227.
- [3] *Microsoft: Project Silica*. URL: <https://www.microsoft.com/en-us/research/project/project-silica/>.
- [4] A Talebpour, S Larochelle, and S L Chin. “Non-sequential and sequential double ionization of NO in an intense femtosecond Ti:sapphire laser pulse”. In: *Journal of Physics B: Atomic, Molecular and Optical Physics* 30.7 (Apr. 1997), pp. L245–L250. DOI: 10.1088/0953-4075/30/7/002.
- [5] A. l’Huillier et al. “Multiple charged ions induced by multiphoton absorption in rare gases at 0.53 μm ”. In: *Phys. Rev. A* 27 (5 May 1983), pp. 2503–2512. DOI: 10.1103/PhysRevA.27.2503.
- [6] E. Eremina et al. “Influence of Molecular Structure on Double Ionization of N₂ and O₂ by High Intensity Ultrashort Laser Pulses”. In: *Phys. Rev. Lett.* 92 (17 Apr. 2004), p. 173001. DOI: 10.1103/PhysRevLett.92.173001.
- [7] E Eremina et al. “Laser-induced non-sequential double ionization investigated at and below the threshold for electron impact ionization”. In: *Journal of Physics B: Atomic, Molecular and Optical Physics* 36.15 (July 2003), pp. 3269–3280. DOI: 10.1088/0953-4075/36/15/308.
- [8] D. Zeidler et al. “Controlling Attosecond Double Ionization Dynamics via Molecular Alignment”. In: *Phys. Rev. Lett.* 95 (20 Nov. 2005), p. 203003. DOI: 10.1103/PhysRevLett.95.203003.
- [9] Yunquan Liu et al. “Multiphoton Double Ionization of Ar and Ne Close to Threshold”. In: *Phys. Rev. Lett.* 104 (17 Apr. 2010), p. 173002. DOI: 10.1103/PhysRevLett.104.173002.
- [10] Boris Bergues et al. “Attosecond tracing of correlated electron-emission in non-sequential double ionization”. In: *Nature Communications* 3.1 (2012), p. 813. ISSN: 2041-1723. DOI: 10.1038/ncomms1807.
- [11] Xufei Sun et al. “Mechanisms of Strong-Field Double Ionization of Xe”. In: *Phys. Rev. Lett.* 113 (10 Sept. 2014), p. 103001. DOI: 10.1103/PhysRevLett.113.103001.
- [12] T. Shaaran, M. T. Nygren, and C. Figueira de Morisson Faria. “Laser-induced nonsequential double ionization at and above the recollision-excitation-tunneling threshold”. In: *Phys. Rev. A* 81 (6 June 2010), p. 063413. DOI: 10.1103/PhysRevA.81.063413.
- [13] Zhangjin Chen, Yaqiu Liang, and C. D. Lin. “Quantitative rescattering theory of correlated two-electron momentum spectra for strong-field nonsequential double ionization of helium”. In: *Phys. Rev. A* 82 (6 Dec. 2010), p. 063417. DOI: 10.1103/PhysRevA.82.063417.
- [14] Jonathan Parker and Charles W. Clark. “Study of a plane-wave final-state theory of above-threshold ionization and harmonic generation”. In: *J. Opt. Soc. Am. B* 13.2 (Feb. 1996), pp. 371–379. DOI: 10.1364/JOSAB.13.000371.

- [15] C Faria and AS Maxwell. “It is all about phases: ultrafast holographic photoelectron imaging”. In: *arXiv preprint arXiv:1906.11781* (2019).
- [16] P. A. Franken et al. “Generation of Optical Harmonics”. In: *Phys. Rev. Lett.* 7 (4 Aug. 1961), pp. 118–119. DOI: 10.1103/PhysRevLett.7.118.
- [17] Ahmed Al-Jawahiry. *Coulomb Focusing and Interference Effects in Above-Threshold Ionization (MSc thesis)*. 2017.
- [18] Andrew S Maxwell. “Strong-Field Interference of Quantum Trajectories with Coulomb Distortion and Electron Correlation (Doctoral thesis)”. University College London, 2018.
- [19] Benjamin Wolter et al. “Strong-Field Physics with Mid-IR Fields”. In: *Phys. Rev. X* 5 (2 June 2015), p. 021034. DOI: 10.1103/PhysRevX.5.021034.
- [20] A. S. Maxwell and C. Figueira de Morisson Faria. “Quantum interference in time-delayed nonsequential double ionization”. In: *Phys. Rev. A* 92 (2 Aug. 2015), p. 023421. DOI: 10.1103/PhysRevA.92.023421.
- [21] Zhangjin Chen, Yaqiu Liang, and C. D. Lin. “Quantum Theory of Recollisional (e , $2e$) Process in Strong Field Nonsequential Double Ionization of Helium”. In: *Phys. Rev. Lett.* 104 (25 June 2010), p. 253201. DOI: 10.1103/PhysRevLett.104.253201.
- [22] M Kübel et al. “Non-sequential double ionization of Ar: from the single- to the many-cycle regime”. In: *New Journal of Physics* 16.3 (Mar. 2014), p. 033008. DOI: 10.1088/1367-2630/16/3/033008.
- [23] Rebecca Jade Tenney. “Controlling Quantum Interference in Nonsequential Double Ionization”. MA thesis. University College London, 2018.
- [24] *Hartree atomic units*. Dec. 2019.
- [25] Lars Madsen. *STRONG FIELD DYNAMICS Scattering and Multiphoton Ionization*.
- [26] A. S. Maxwell and C. Figueira de Morisson Faria. “Controlling Below-Threshold Nonsequential Double Ionization via Quantum Interference”. In: *Phys. Rev. Lett.* 116 (14 Apr. 2016), p. 143001. DOI: 10.1103/PhysRevLett.116.143001.
- [27] Dieter Bauer and Peter Koval. “Qprop: A Schrödinger-solver for intense laser-atom interaction”. In: *Computer physics communications* 174.5 (2006), pp. 396–421.
- [28] J. S. Parker et al. “High-Energy Cutoff in the Spectrum of Strong-Field Nonsequential Double Ionization”. In: *Phys. Rev. Lett.* 96 (13 Apr. 2006), p. 133001. DOI: 10.1103/PhysRevLett.96.133001.
- [29] Misha Yu Ivanov, Michael Spanner, and Olga Smirnova. “Anatomy of strong field ionization”. In: *Journal of Modern Optics* 52.2-3 (2005), pp. 165–184. DOI: 10.1080/0950034042000275360.
- [30] Tahir Shaaran. “A rigorous treatment of excitation and quantum interference in laser-induced nonsequential double ionization of atoms and molecules”. PhD thesis. University College London, 2011.
- [31] Wilhelm Becker et al. “Theories of photoelectron correlation in laser-driven multiple atomic ionization”. In: *Rev. Mod. Phys.* 84 (3 July 2012), pp. 1011–1043. DOI: 10.1103/RevModPhys.84.1011.

- [32] Tian-Min Yan et al. “Low-Energy Structures in Strong Field Ionization Revealed by Quantum Orbits”. In: *Phys. Rev. Lett.* 105 (25 Dec. 2010), p. 253002. DOI: 10.1103/PhysRevLett.105.253002.
- [33] Kasra Amini et al. “Symphony on Strong Field Approximation”. In: *Reports on Progress in Physics* (2019). DOI: 10.1088/1361-6633/ab2bb1.
- [34] Maria Göppert-Mayer. “Über Elementarakte mit zwei Quantensprüngen”. In: *Annalen der Physik* 401.3 (1931), pp. 273–294. DOI: 10.1002/andp.19314010303.
- [35] Armin Scrinzi. *Attosecond Physics -Theory Lecture notes 2011 (Work in progress)*. 2011.
- [36] W. Gordon. “Der Comptoneffekt nach der Schrödingerschen Theorie”. In: *Zeitschrift für Physik* 40.1 (Jan. 1926), pp. 117–133. ISSN: 0044-3328. DOI: 10.1007/BF01390840.
- [37] W. Becker et al. “Above-Threshold Ionization: From Classical Features to Quantum Effects”. In: ed. by Benjamin Bederson and Herbert Walther. Vol. 48. *Advances In Atomic, Molecular, and Optical Physics*. Academic Press, 2002, pp. 35–98. DOI: [https://doi.org/10.1016/S1049-250X\(02\)80006-4](https://doi.org/10.1016/S1049-250X(02)80006-4).
- [38] G.B. Arfken and H.J Weber. *Mathematical Methods for Theoretical Physicists (7th edition)*. Academic Press, 2012.
- [39] Kenneth G. Kay. “Time-dependent semiclassical tunneling through barriers”. In: *Physical Review A* 88.1 (July 2013). DOI: 10.1103/physreva.88.012122.
- [40] P. B. Corkum. “Plasma perspective on strong field multiphoton ionization”. In: *Phys. Rev. Lett.* 71 (13 Sept. 1993), pp. 1994–1997. DOI: 10.1103/PhysRevLett.71.1994.
- [41] Dieter Bauer, DB Milošević, and W Becker. “Strong-field approximation for intense-laser-atom processes: The choice of gauge”. In: *Physical Review A* 72.2 (2005), p. 023415. DOI: 10.1103/PhysRevA.72.023415.
- [42] Olga Smirnova, Michael Spanner, and Mikhail Ivanov. “Anatomy of strong field ionization II: To dress or not to dress?” In: *Journal of Modern Optics* 54 (May 2007). DOI: 10.1080/09500340701234656.
- [43] X.-Y. Lai et al. “Influence of the Coulomb potential on above-threshold ionization: A quantum-orbit analysis beyond the strong-field approximation”. In: *Phys. Rev. A* 92 (4 Oct. 2015), p. 043407. DOI: 10.1103/PhysRevA.92.043407.
- [44] S. V. Popruzhenko. “Invariant form of Coulomb corrections in the theory of nonlinear ionization of atoms by intense laser radiation”. In: *Journal of Experimental and Theoretical Physics* 118.4 (Apr. 2014), pp. 580–586. ISSN: 1090-6509. DOI: 10.1134/S1063776114040062.
- [45] A. S. Maxwell, S. V. Popruzhenko, and C. Figueira de Morisson Faria. “Treating branch cuts in quantum trajectory models for photoelectron holography”. In: *Phys. Rev. A* 98 (6 Dec. 2018), p. 063423. DOI: 10.1103/PhysRevA.98.063423.
- [46] Emilio Pisanty and Misha Ivanov. “Slalom in complex time: Emergence of low-energy structures in tunnel ionization via complex-time contours”. In: *Physical Review A* 93.4 (Apr. 2016). DOI: 10.1103/physreva.93.043408.

- [47] A S Maxwell et al. “Analytic quantum-interference conditions in Coulomb corrected photoelectron holography”. In: *Journal of Physics B: Atomic, Molecular and Optical Physics* 51.4 (Jan. 2018), p. 044004. DOI: 10.1088/1361-6455/aa9e81.
- [48] D. B. Milošević. “Semiclassical approximation for strong-laser-field processes”. In: *Phys. Rev. A* 96 (2 Aug. 2017), p. 023413. DOI: 10.1103/PhysRevA.96.023413.
- [49] Prof. Carla Figueira De Morisson Faria. *UCL Course PHAS0069: Advanced Quantum Theory Lecture Notes*. Sept. 2019.
- [50] T. Shaaran, M. T. Nygren, and C. Figueira de Morisson Faria. “Laser-induced nonsequential double ionization at and above the recollision-excitation-tunneling threshold”. In: *Physical Review A* 81.6 (June 2010). DOI: 10.1103/physreva.81.063413.
- [51] C. Figueira de Morisson Faria, H. Schomerus, and W. Becker. “High-order above-threshold ionization: The uniform approximation and the effect of the binding potential”. In: *Physical Review A* 66.4 (Oct. 2002). DOI: 10.1103/physreva.66.043413.
- [52] D. B. Milošević and W. Becker. “Classical cutoffs for laser-induced nonsequential double ionization”. In: *Physical Review A* 68.6 (Dec. 2003). DOI: 10.1103/physreva.68.065401.
- [53] XiaoLei Hao et al. “Quantum Effects in Double Ionization of Argon below the Threshold Intensity”. In: *Phys. Rev. Lett.* 112 (7 Feb. 2014), p. 073002. DOI: 10.1103/PhysRevLett.112.073002.
- [54] T. Shaaran, K. Z. Hatsagortsyan, and C. H. Keitel. “Coulomb effect in laser-induced recollision excitation”. In: *Phys. Rev. A* 98 (2 Aug. 2018), p. 023410. DOI: 10.1103/PhysRevA.98.023410.
- [55] T. Shaaran et al. “Role of high ponderomotive energy in laser-induced nonsequential double ionization”. In: *Phys. Rev. A* 99 (2 Feb. 2019), p. 023421. DOI: 10.1103/PhysRevA.99.023421.
- [56] G. P. Katsoulis et al. “Slingshot Nonsequential Double Ionization as a Gate to Anticorrelated Two-Electron Escape”. In: *Phys. Rev. Lett.* 121 (26 Dec. 2018), p. 263203. DOI: 10.1103/PhysRevLett.121.263203.
- [57] Prof. Timo Betcke. *UCL Course PHAS0102: Techniques in High-Performance-Computing*. Dec. 2019.
- [58] Vancea Codruța, Vancea Florin, and Nicula Antoniu. *On Double Interpolation in Polar Coordinates*. https://www.researchgate.net/publication/40422444_On_Double_Interpolation_in_Polar_Coordinates. Oct. 2009.
- [59] Borislav Bojanov and Yuan Xu. “On polynomial interpolation of two variables”. In: *Journal of Approximation Theory* 120.2 (Feb. 2003), pp. 267–282. DOI: 10.1016/s0021-9045(02)00023-0.
- [60] Georg Muntingh. *Topics in Polynomial Interpolation Theory (PhD thesis)*. <https://core.ac.uk/download/pdf/30899611.pdf>. 2010.
- [61] *The Keldysh Ionization Parameter*. 2010. URL: http://www.iuac.res.in/atmol/~safvan/safvan_thesis/node32.html.

- [62] Lisa Torlina and Olga Smirnova. “Time-dependent analytical R-matrix approach for strong-field dynamics in One-electron systems”. In: *Physical Review A* 86.4 (Oct. 2012). DOI: 10.1103/physreva.86.043408.